Western and Central Pacific Fisheries Commission

Workflow demonstration

Assessment workflow demo: ALB

WORKFLOW DEMO

2026-09-05

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1 Executive summary

This is a workflow demonstration, not a stock assessment. It uses ALB outputs to demonstrate the model-to-report tools. The diagnostic-model figures use the archived MFCL-compatible KIMSA fit from job 24781. Stepwise compares completed native MFCL runs (INI 1003 and INI 1007) with the same KIMSA baseline. Starting-value, self-test, retrospective, sensitivity, ensemble, profile, projection, hindcast and additional-output figures use illustrative display values to demonstrate report construction. They are not new assessment estimates.

Table 1 links the report and all interactive supplements. Replace the illustrative study bundles with fitted results, write the assessment interpretation in the section files, and update the report cover before submission.

2 Introduction

3 Data compilation

The diagnostic model retains the ALB observation timing and two-region structure. Catch, abundance-index, length-composition and conditional age-at-length observations are available in the interactive diagnostic supplement.

Region boundaries follow the 2024 South Pacific albacore assessment, Rev.03, Section 4.1 and Figure 2. The spatial comparison also shows its Figure 1 reproduction of the 2021 structure.

Table 2 describes the fishery roles and observation periods. Figure 6 shows data availability. Index observations and predictions follow the model likelihood's geometric-mean centering convention.

4 Model description

The archived ALB fit uses the MFCL-compatible age-region state, age-based selectivity and five-pass hybrid F, without tagging. Its original F upper bound of 3 is retained for reproduction; new model specifications default to 6.

Length compositions use square_fita with fishery divisors and a raw-record sample-size cap of 1000.

Figure 13, Figure 14 and Figure 15 describe the biological inputs and estimates; Figure 40 shows fishery selectivity. Additional-format panels illustrate output types that are not estimated by this tag-free ALB reference model.

5 Model development

Figure 16 compares the latest completed matching ALB baselines found in Kflow: native MFCL 2.2.7.0 / INI 1003 (job 24514), native MFCL 2.2.7.9 / INI 1007 (job 24515), then KIMSA / hybrid F (job 24781). All used TunaFlow 2.5. These panels use the archived annual report censuses; they are distinct from the time-weighted annual summaries of the detailed diagnostic snapshot.

6 Results

Figure 39 shows annual reference-model trajectories. Stocks use time-weighted annual means; recruitment and catches use annual totals. Depletion is the ratio of annual spawning-potential summaries. Index points and predictions use annual geometric means. The interactive viewer opens annual summaries; select Original to inspect the fitted time steps.

Length-residual panels retain the original fitted records and their compressed-tail support. Conditional age-at-length fits are pooled by region using raw sample counts, preserving the model's prediction scale.

Index fits and residuals (Figure 18), length-composition residuals and conditional age-at-length fits (Figure 26) are generated from the same result bundle used by the interactive viewer.

The following studies demonstrate the intended diagnostic presentation: starting-value trials (Figure 27), simulation-estimation recovery (Figure 30), retrospective analysis (Figure 32), objective profiles (Figure 33) and sensitivity comparisons. Their displayed values are illustrative. Hindcasting (Figure 73; Table 14) exercises withheld-data prediction with inexpensive log-linear index trends, not refitted stock-assessment models.

7 Reporting examples

The ensemble and management-quantity panels demonstrate the reporting formats. Projection panels compare fixed-catch and F scenarios using a simple surplus-production kernel initialized at the ALB biomass scale; its dynamics and variability are illustrative. They do not establish ALB stock status. Supply actual reference points, model retention decisions, uncertainty draws and projection results before writing management conclusions.

8 Discussion

9 Tables

Table 1: Public report, interactive supplements and published outputs for this workflow demonstration.

Resource	Figures and tables	Interactive output
Workflow demonstration	PDF report	HTML report
Diagnostic fit	HTML supplement	Model viewer
Stepwise development	HTML supplement	Model viewer
Jitter	HTML supplement	Model viewer
Self-test	HTML supplement	Model viewer
Retrospective analysis	HTML supplement	Model viewer
Sensitivities	HTML supplement	Model viewer
Uncertainty ensemble	HTML supplement	Model viewer
Objective profiles	HTML supplement	Model viewer
Additional output examples	HTML supplement	Model viewer
Hindcasting	HTML supplement	Model viewer
Projection scenarios	HTML supplement	Model viewer
Published outputs	Public repository	Rendered artifacts

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t

Fishery	Region	Role	First year	Last year	Unit
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Illustrative output. Dirichlet–multinomial composition settings.

Fishery	Likelihood	Group	Theta
Fishery A	dirichlet_multinomial	1	20

Table 4: Illustrative output. Groups sharing composition-weight parameters.

Group	Fishery
1	Fishery A

Table 5: Illustrative output. Tag-release and reporting groups.

Programme	Region	Mixing
Programme A	Region 1	Two supplied model periods

Table 6: Illustrative output. Parameters on or near their declared bounds.

Parameter	Estimate	Lower	Upper
Steepness	0.99	0.2	0.99

Table 7: Illustrative output. Parameter correlations from the supplied covariance estimate.

Parameter	Partner	Correlation
log R0	Index q	-0.97
Initial abundance	Recruitment scale	0.96

Table 8: Management quantities and supplied uncertainty summaries.

Quantity	Period	Median	50% interval	80% interval	95% interval
$SB/SB_{F=0}$	2016–2020	0.45	0.41–0.49	0.37–0.53	0.33–0.57
F/F_{MSY}	2016–2020	0.68	0.61–0.75	0.54–0.82	0.46–0.90
SB/SB_{MSY}	2016–2020	1.35	1.22–1.48	1.08–1.62	0.94–1.76

Table 9: Spawning-depletion comparisons for the declared reference periods.

Period	Reference	Ratio
2016–2020	2012–2015	0.95

Table 10: Stock-status probabilities for the declared reference points.

Condition	Probability
$SB/SB_{F=0} < 0.20$	0.01
$F/F_{MSY} > 1$	0.05
$SB/SB_{MSY} < 1$	0.04

Table 11: Model reference points and their definitions.

Quantity	Median	Lower	Upper	Unit
MSY	60000	48000	72000	t/year
F_{MSY}	0.3	0.24	0.36	per year
SB_{MSY}	1000000	800000	1200000	t
SB_{MSY}/SB_0	0.28	0.22	0.34	ratio

Table 12: Comparison of supplied uncertainty summaries.

Source	Quantity	Median	Lower	Upper
Structural	$SB/SB_{F=0}$	0.45	0.39	0.51
Structural	F/F_{MSY}	0.68	0.57	0.79
Structural and estimation	$SB/SB_{F=0}$	0.45	0.35	0.55
Structural and estimation	F/F_{MSY}	0.68	0.51	0.85

Table 13: Illustrative projected quantities in the final projection year; medians and central 80% intervals.

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	biomass depletion	2052	0.9499	0.9473	0.952	ratio
fixed-catch	catch	2052	45150	45150	45150	t
fixed-catch	fishing mortality	2052	0.01518	0.01514	0.01522	per year
fixed-catch	total biomass	2052	2997000	2989000	3004000	t
higher-F	biomass depletion	2052	0.7427	0.7378	0.7519	ratio
higher-F	catch	2052	180200	179000	182400	t
higher-F	fishing mortality	2052	0.08	0.08	0.08	per year
higher-F	total biomass	2052	2343000	2328000	2372000	t
lower-F	biomass depletion	2052	0.8678	0.8609	0.8719	ratio
lower-F	catch	2052	107400	106500	107900	t
lower-F	fishing mortality	2052	0.04	0.04	0.04	per year
lower-F	total biomass	2052	2738000	2716000	2751000	t

Table 14: Out-of-sample prediction bias and RMSE by series and forecast lead.

Series	Lead	N	Bias	RMSE	Unit
Index 18 (Region 1)	1	3	-0.10032	0.11429	geometric-mean-scaled index
Index 19 (Region 1)	1	3	0.051184	0.13396	geometric-mean-scaled index
Index 20 (Region 2)	1	3	-0.30428	0.34243	geometric-mean-scaled index
Index 18 (Region 1)	2	3	-0.11978	0.1464	geometric-mean-scaled index

Series	Lead	N	Bias	RMSE	Unit
Index 19 (Region 1)	2	3	-0.14261	0.17752	geometric-mean-scaled index
Index 20 (Region 2)	2	3	-0.338	0.43347	geometric-mean-scaled index
Index 18 (Region 1)	3	3	-0.14713	0.152	geometric-mean-scaled index
Index 19 (Region 1)	3	3	0.010767	0.12538	geometric-mean-scaled index
Index 20 (Region 2)	3	3	-0.42539	0.46732	geometric-mean-scaled index

10 Figures

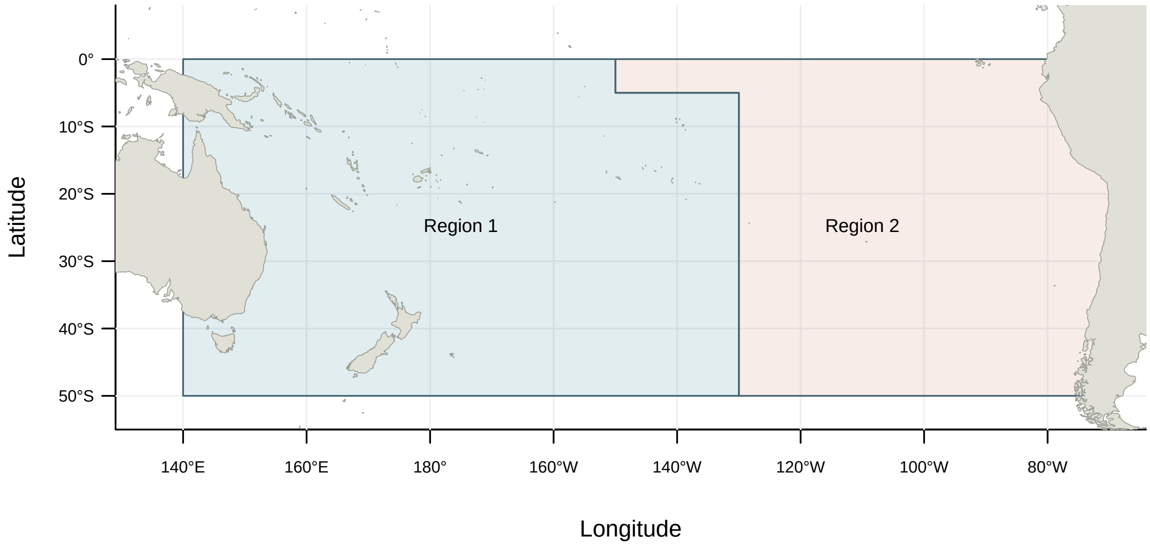

Figure 1: Model regions for the 2024 South Pacific albacore assessment (WCPFC-SC20-SA-WP-02, Rev.03, Figure 2).

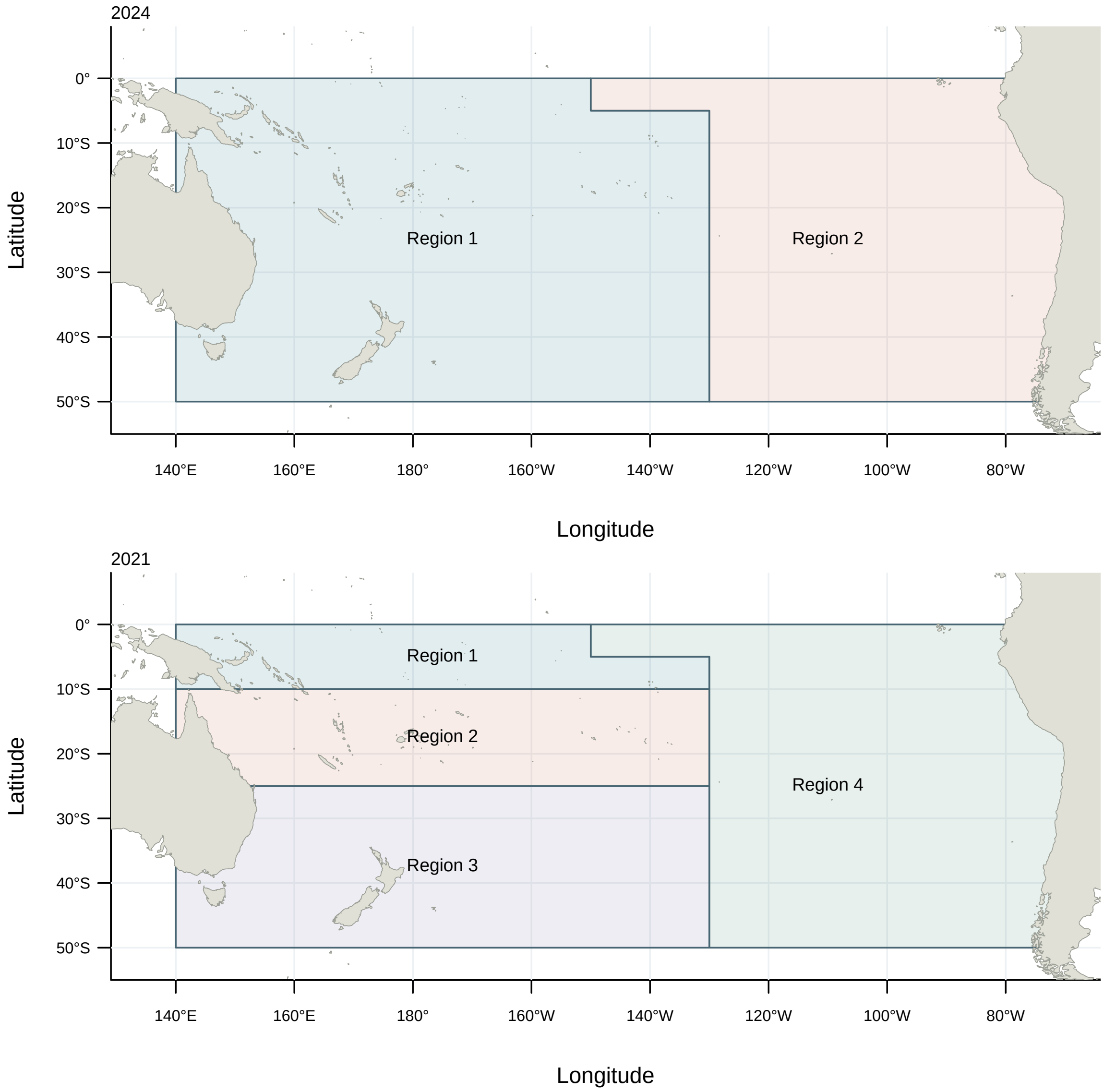

Figure 2: Spatial structures of the 2024 and 2021 South Pacific albacore assessments (WCPFC-SC20-SA-WP-02, Rev.03, Figures 1–2).

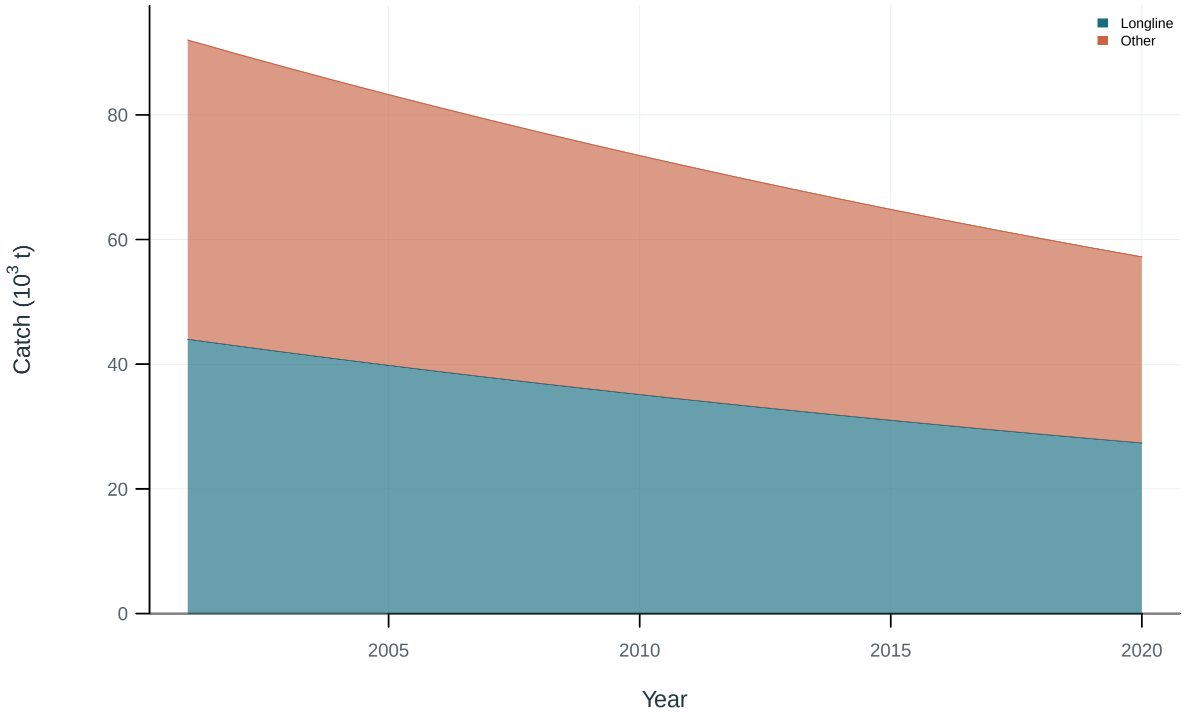

Figure 3: Illustrative output. Catch by fishing group.

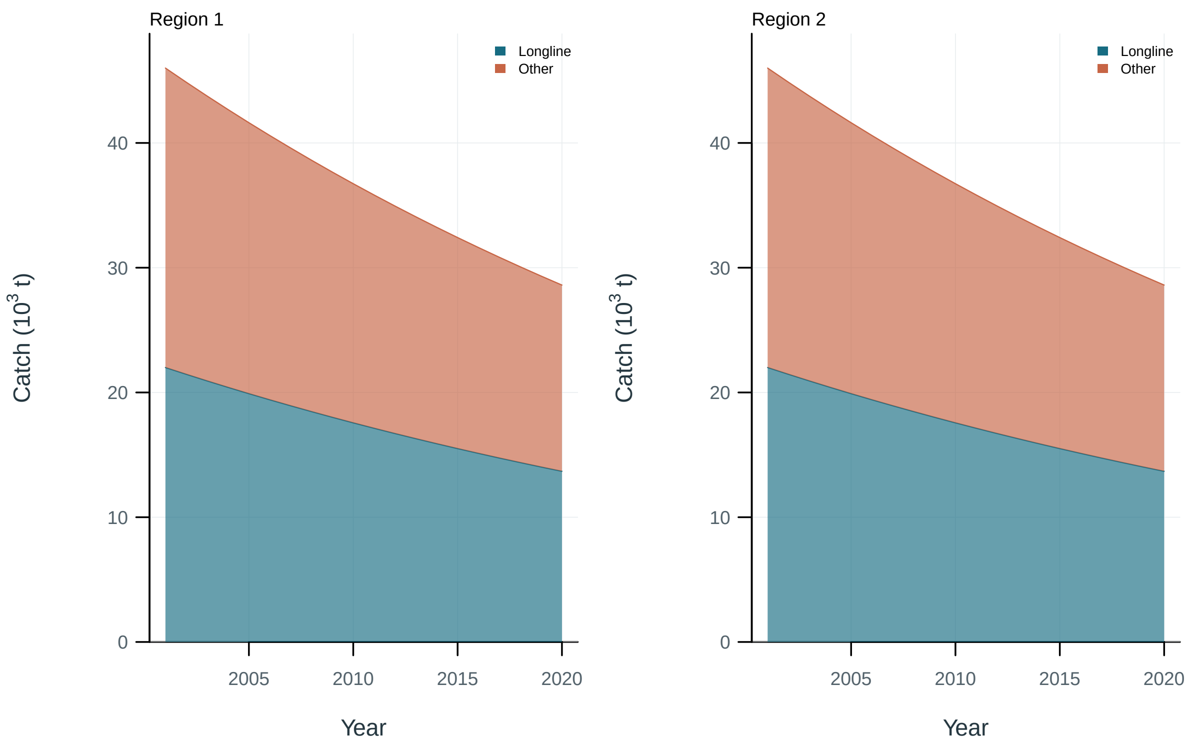

Figure 4: Illustrative output. Catch by fishing group and region.

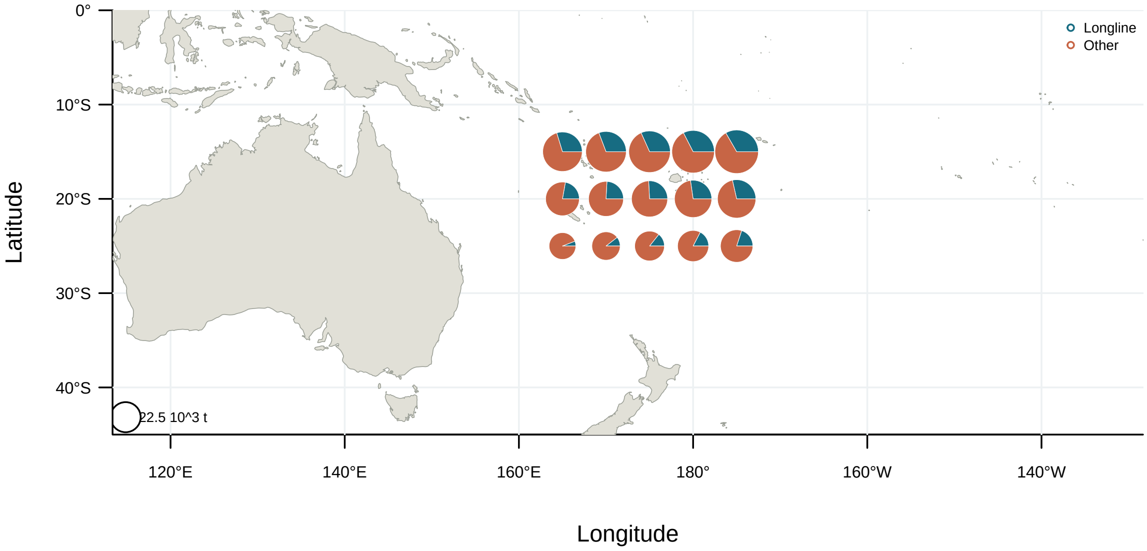

Figure 5: Illustrative output. Spatial catch distribution by fishing group.

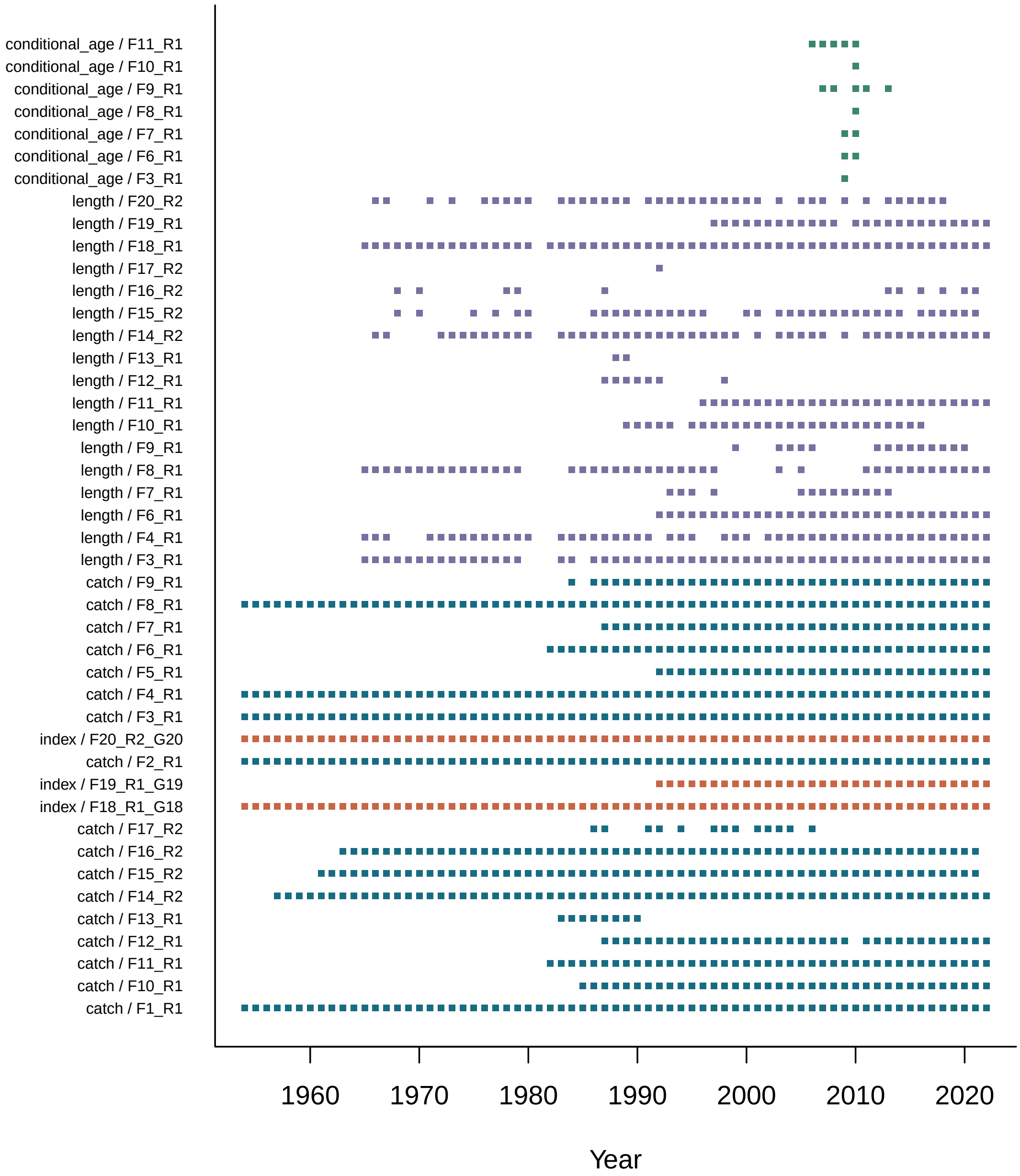

Figure 6: Availability of catch, abundance-index and composition observations by series and year.

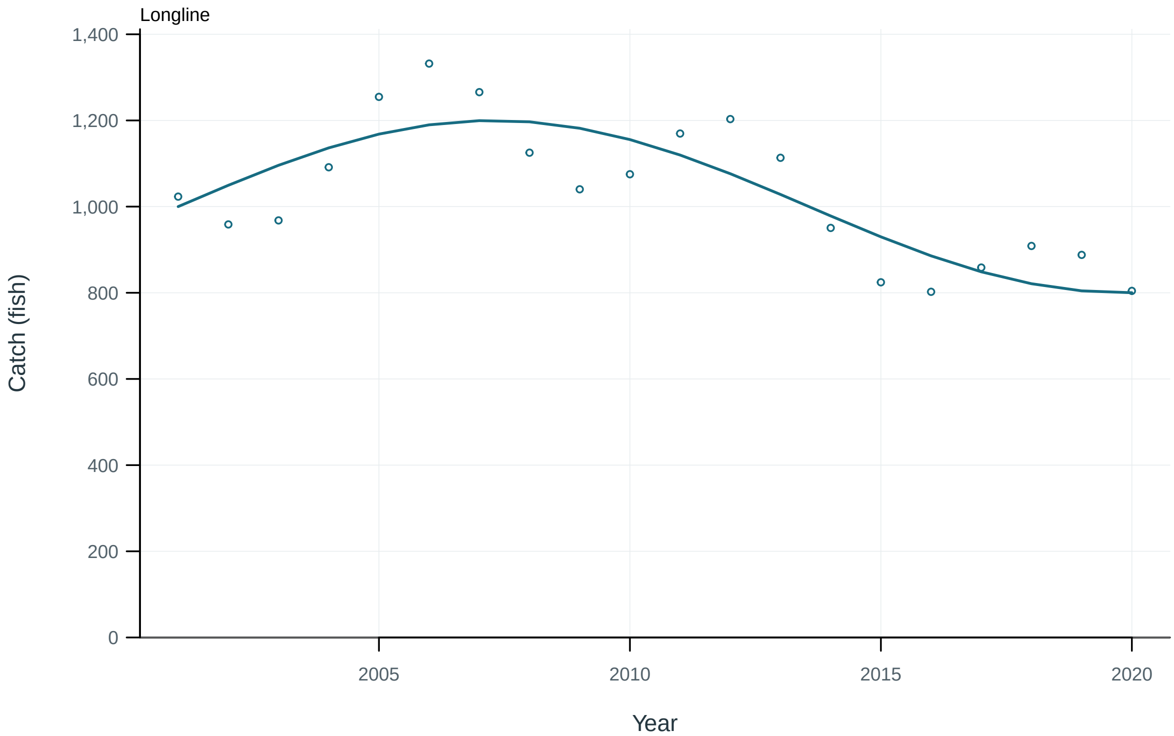

Figure 7: Illustrative output. Observed and predicted catch by fishery.

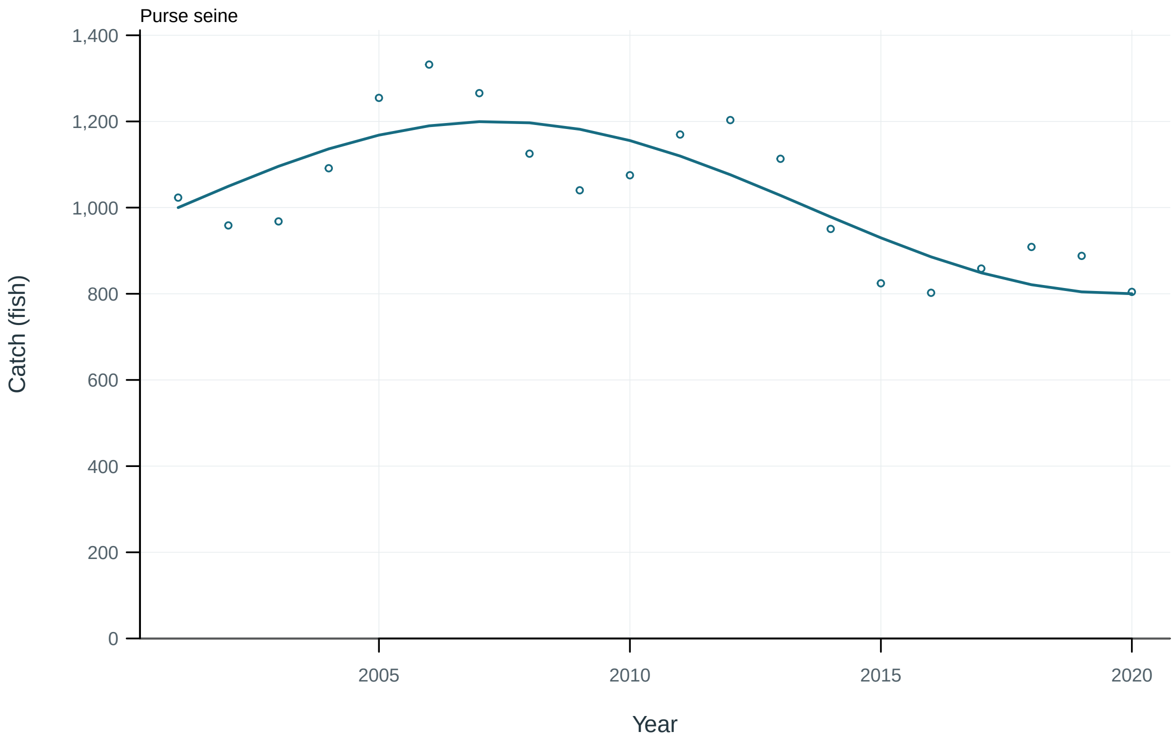

Figure 8: Illustrative output. Observed and predicted catch by fishery.

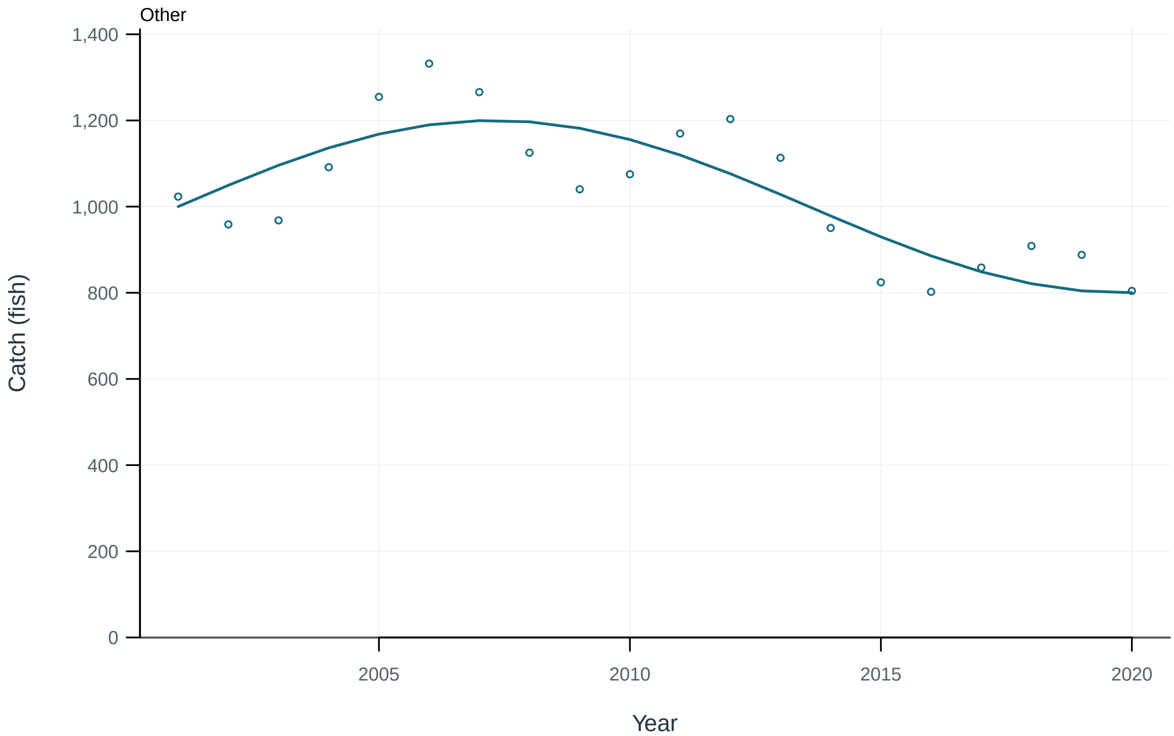

Figure 9: Illustrative output. Observed and predicted catch by fishery.

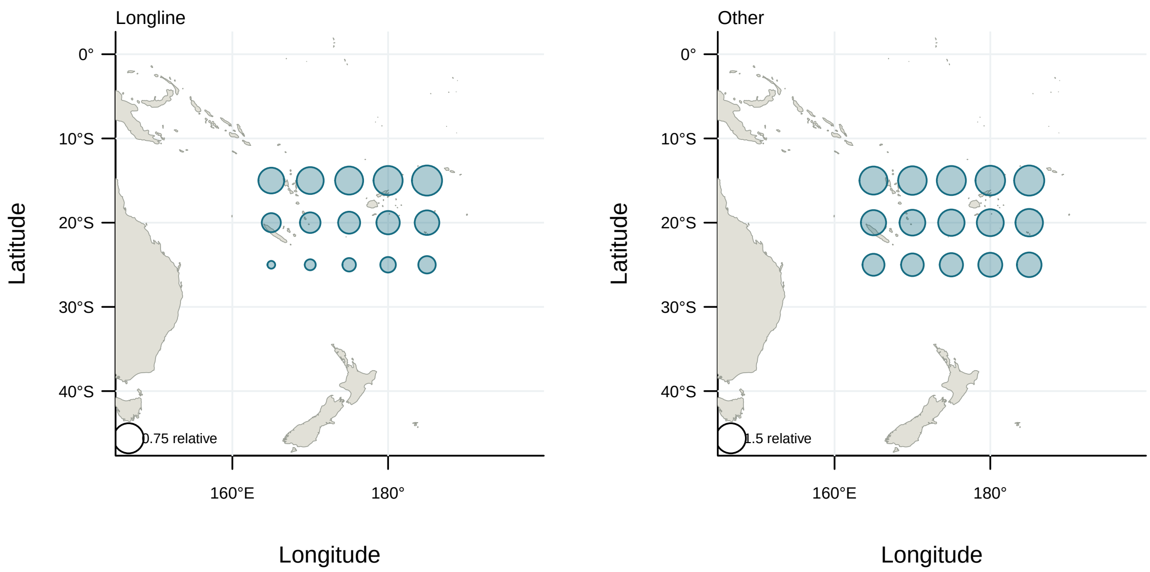

Figure 10: Illustrative output. Spatial distribution of nominal abundance indices.

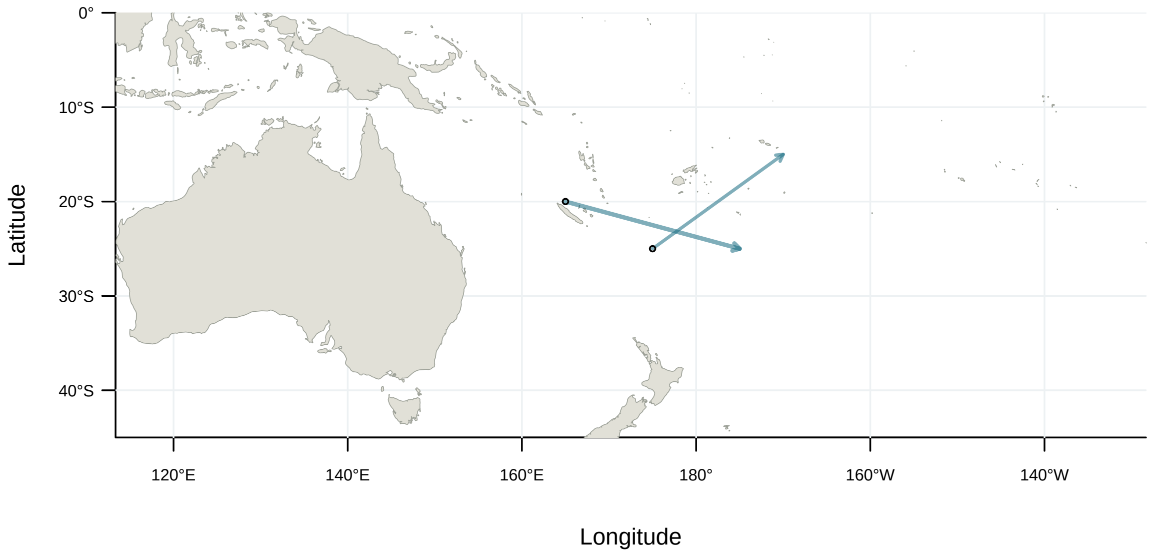

Figure 11: Illustrative output. Tag-release and recapture displacements by programme.

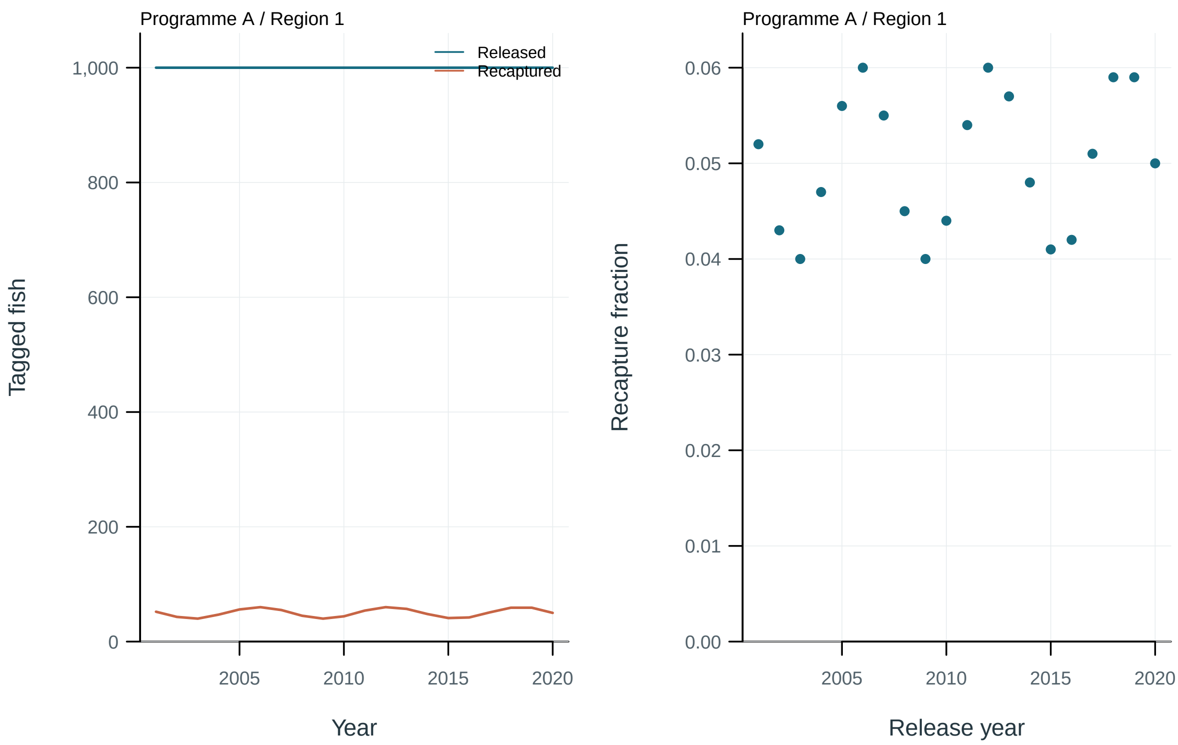

Figure 12: Illustrative output. Tag releases, recaptures and recapture fractions by cohort.

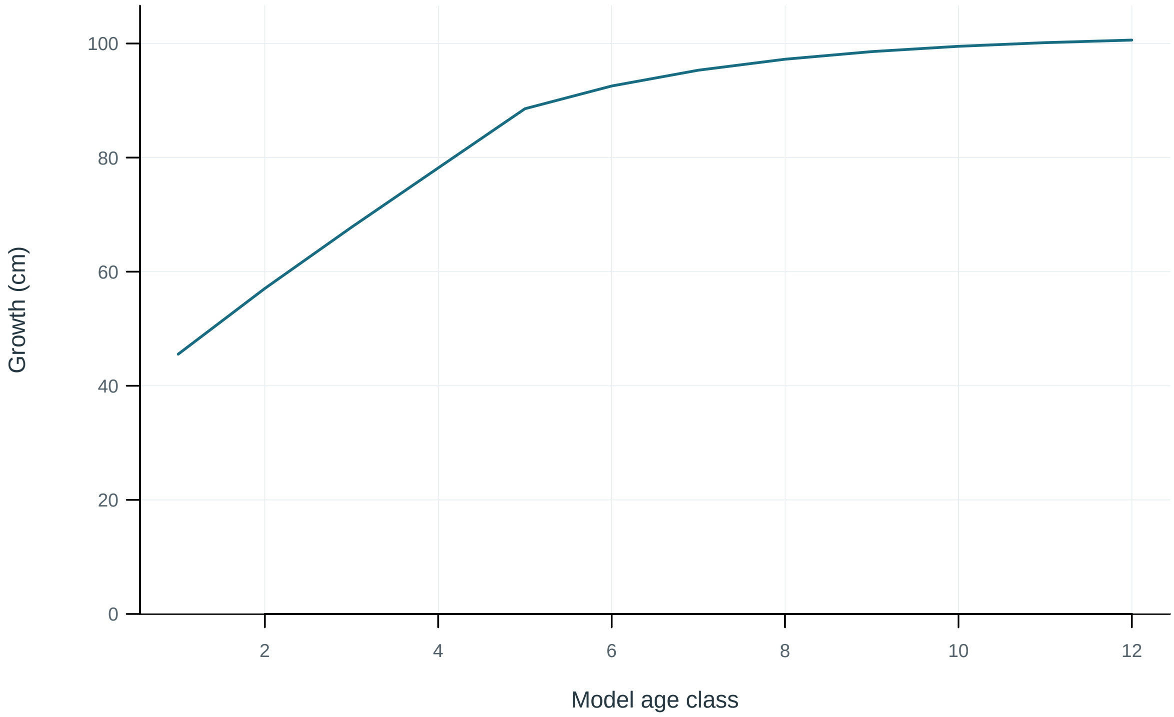

Figure 13: Mean length at the start of each model age class.

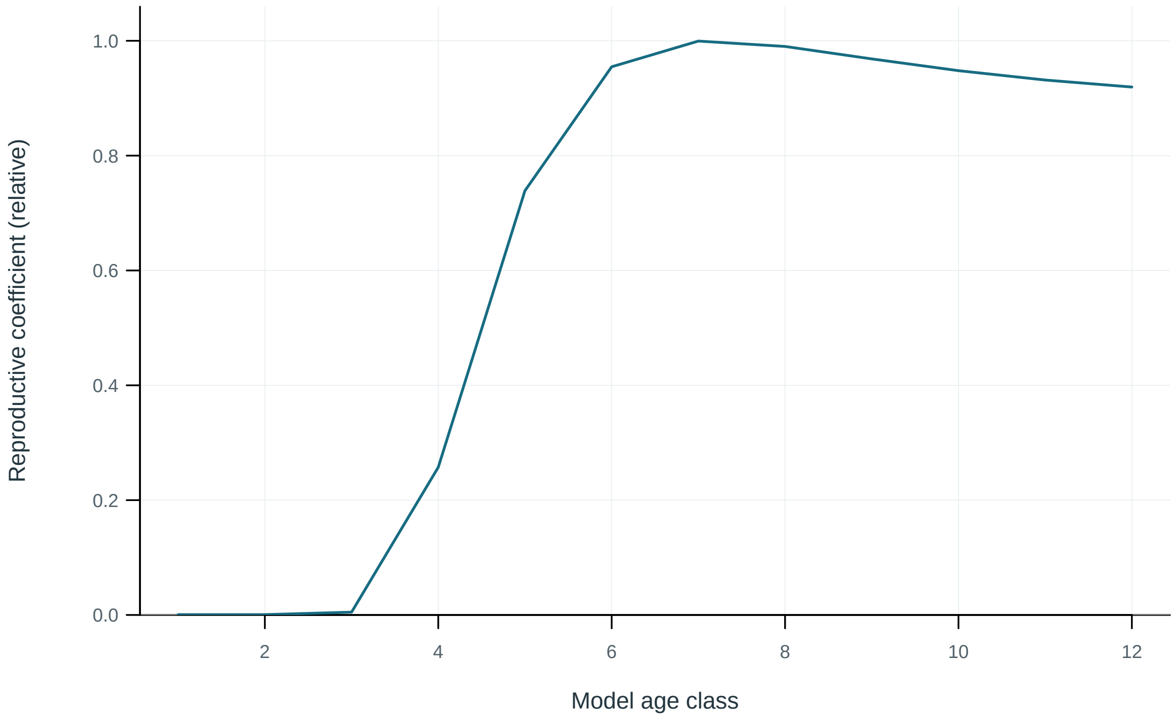

Figure 14: Model reproductive coefficient at age.

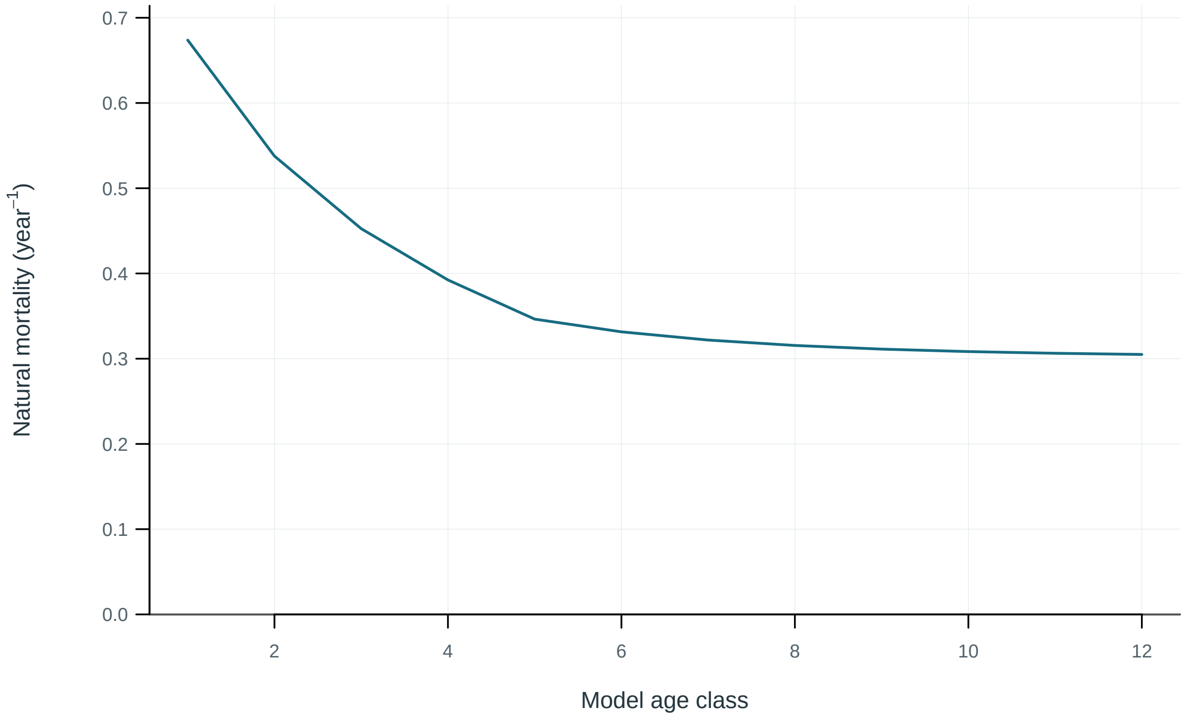

Figure 15: Annual instantaneous natural mortality at age.

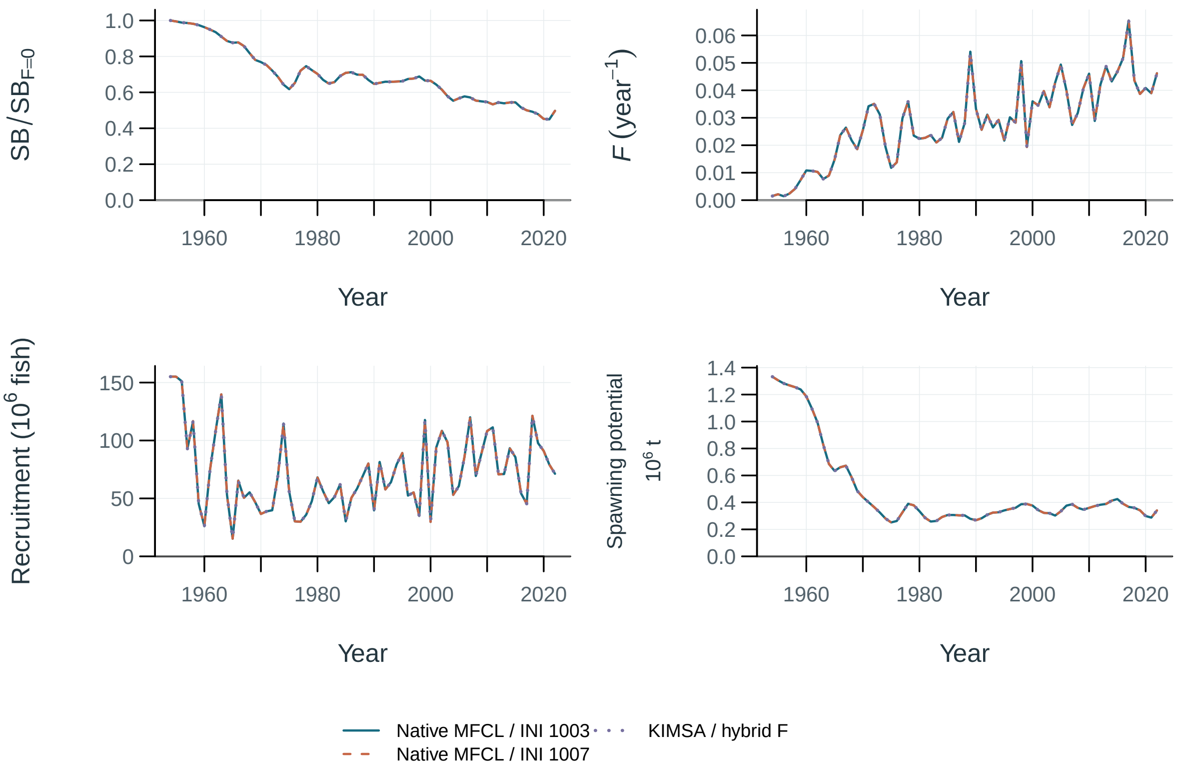

Figure 16: Stepwise model trajectories.

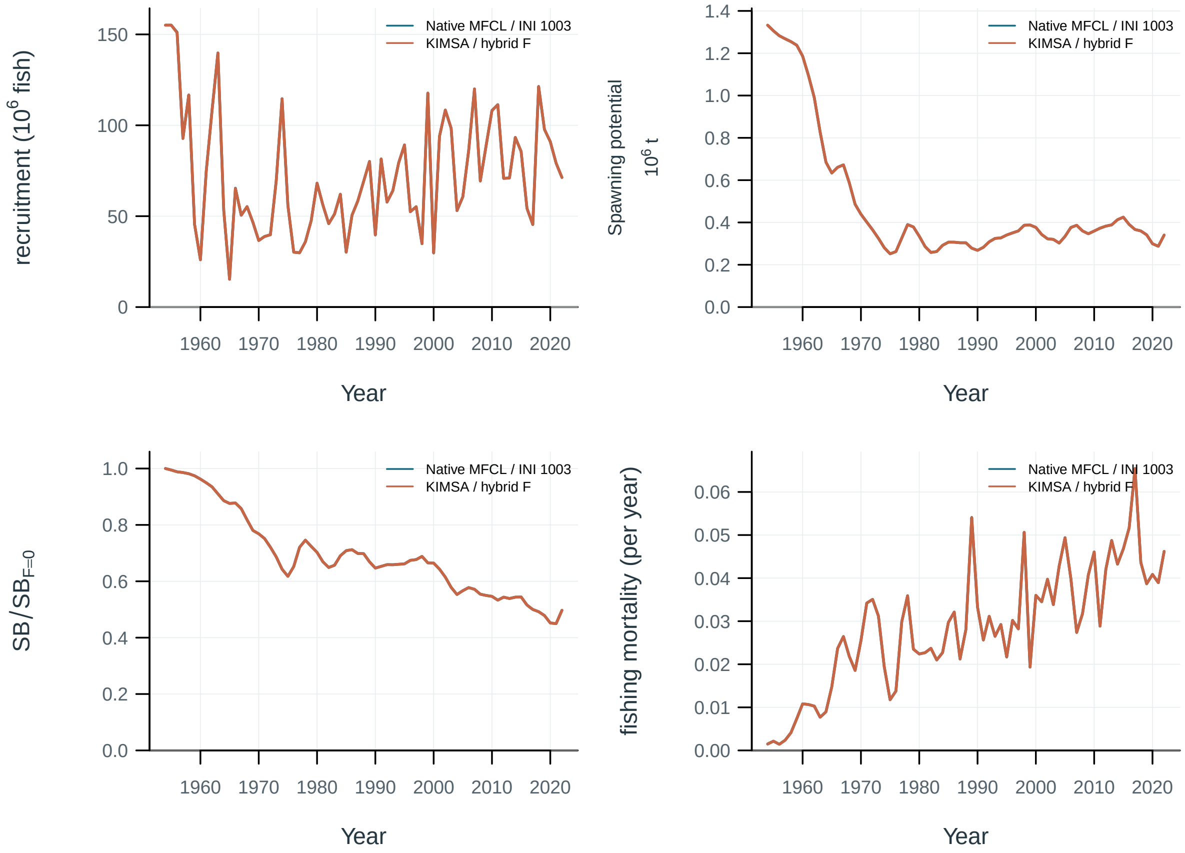

Figure 17: Selected model-development trajectories.

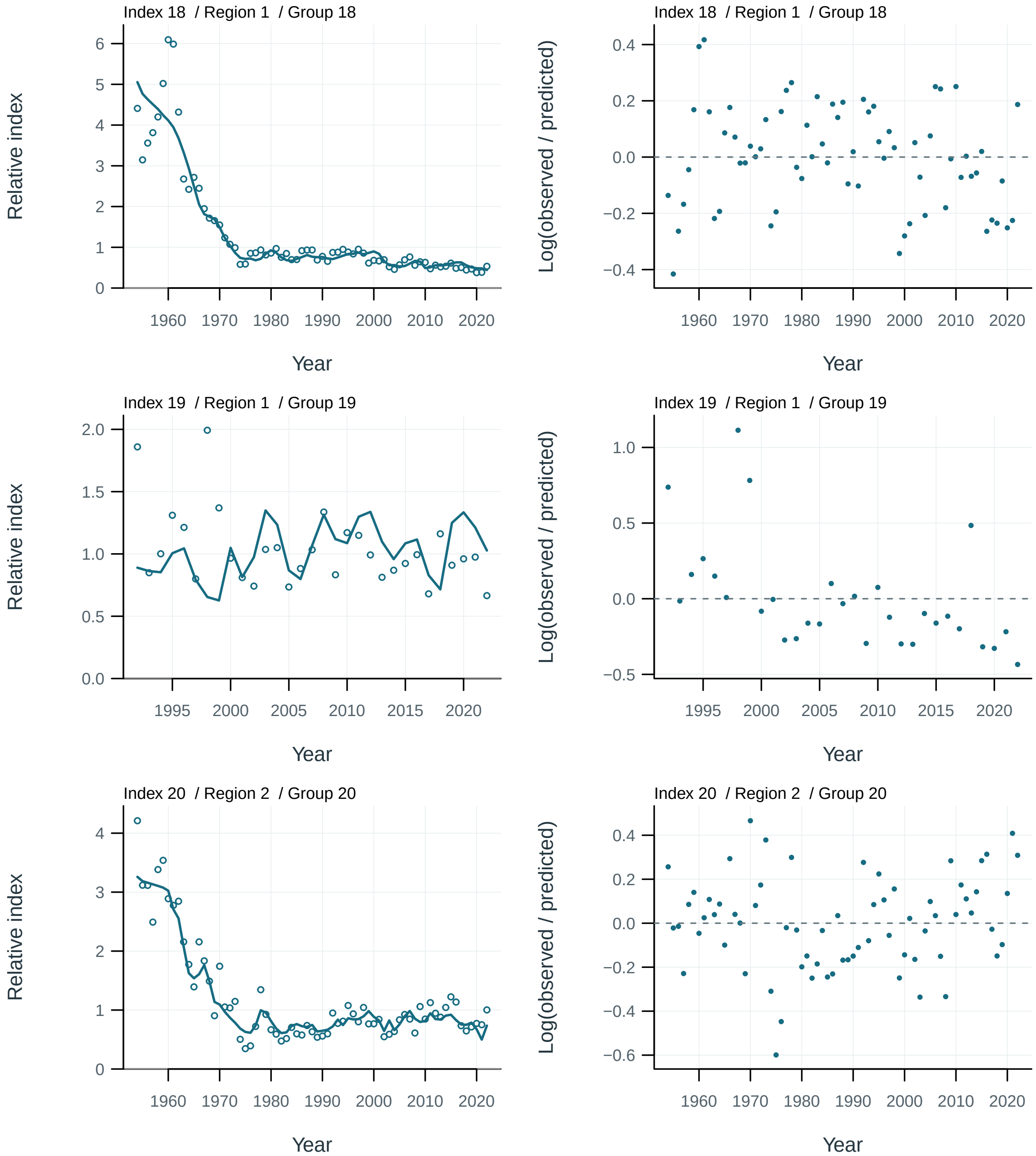

Figure 18: Abundance-index fits and log residuals by series. Residuals are log(observed/predicted).

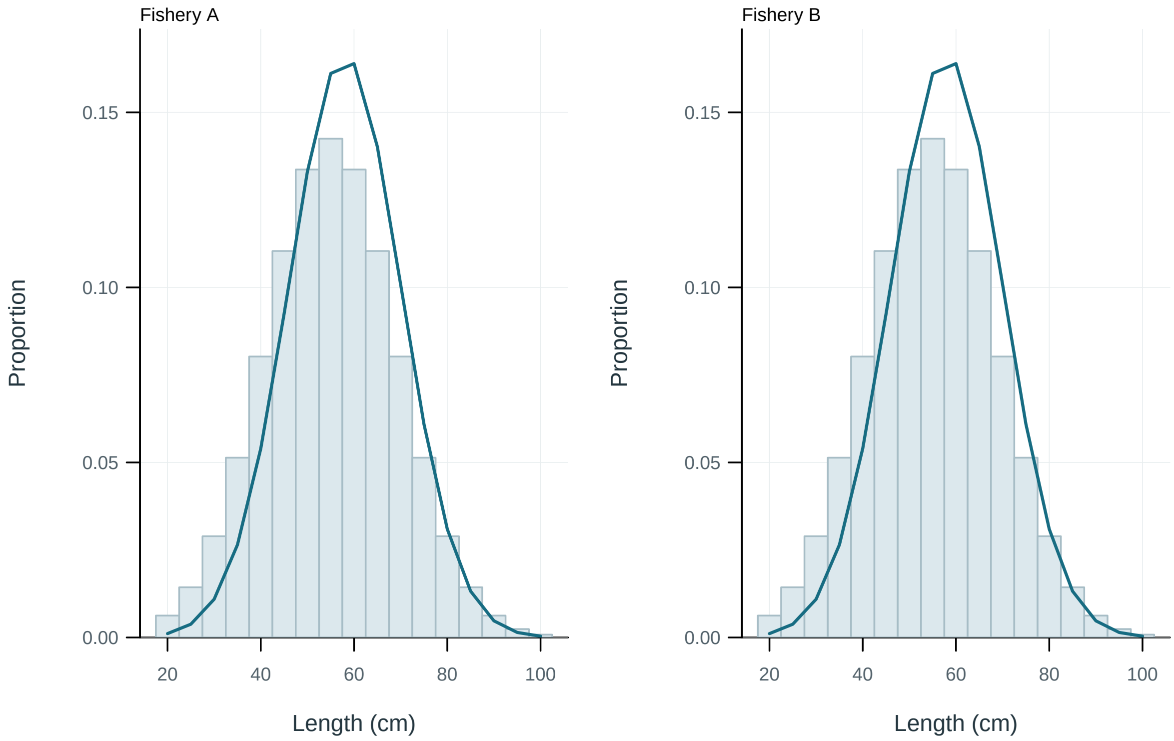

Figure 19: Illustrative output. Pooled observed and predicted length distributions.

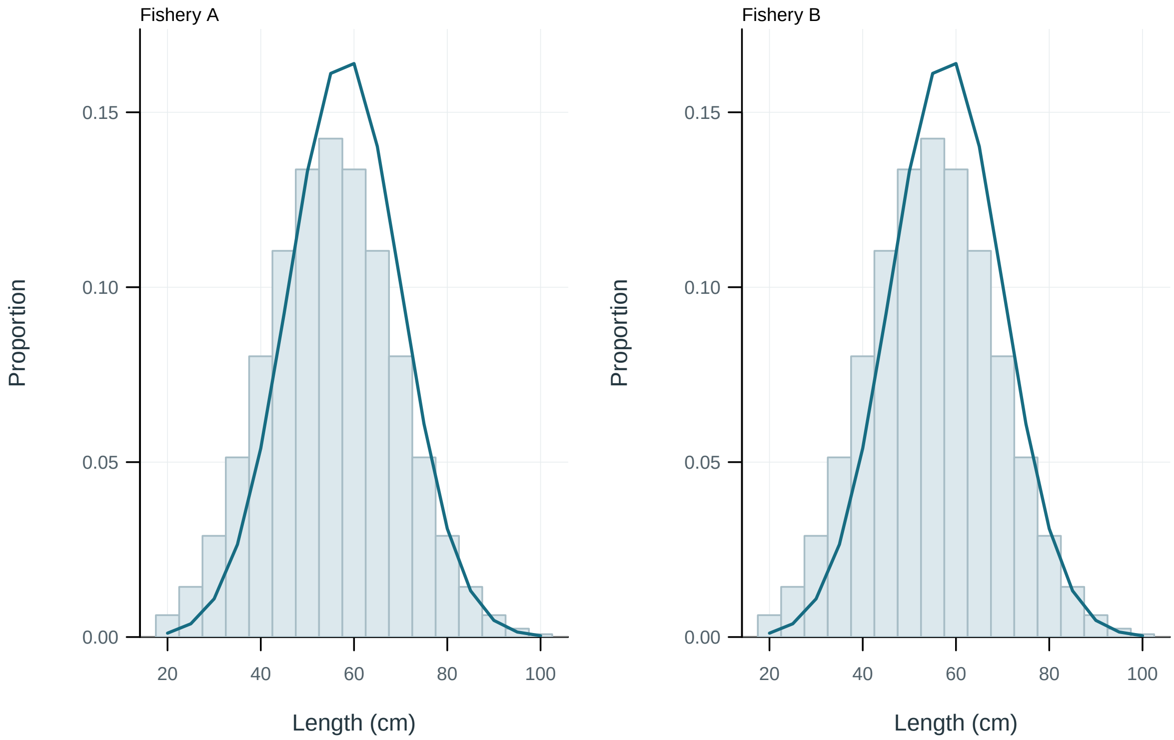

Figure 20: Illustrative output. Pooled observed and predicted length distributions.

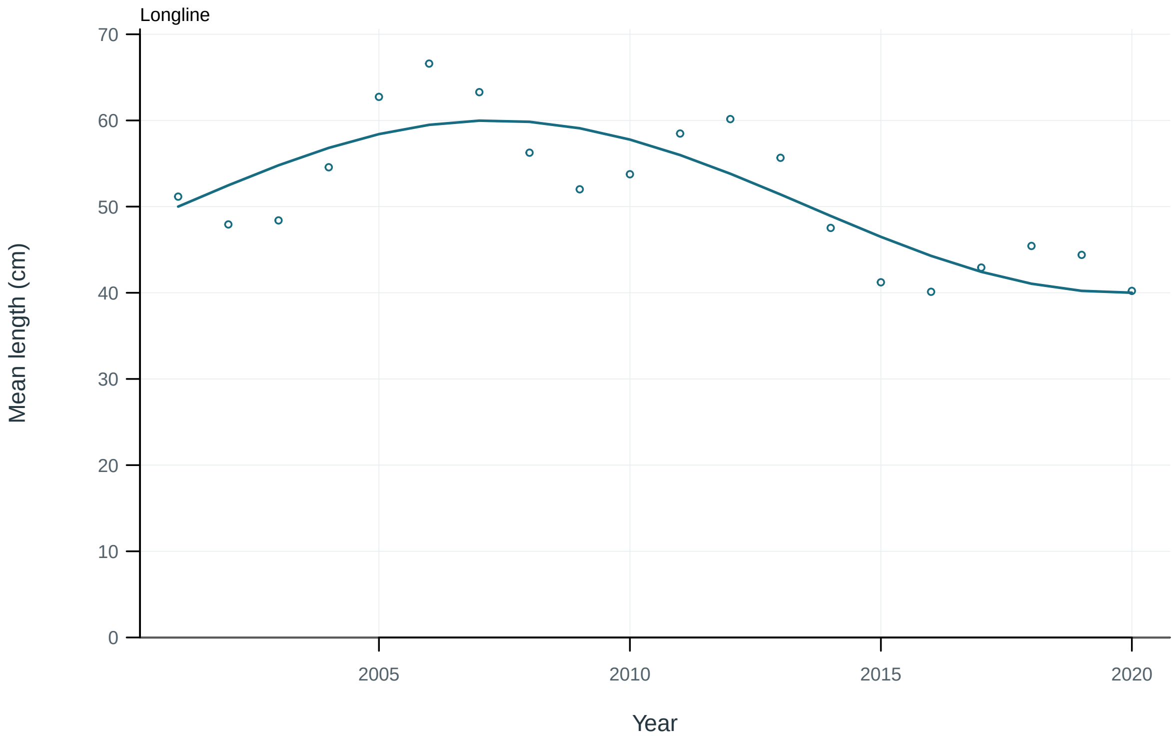

Figure 21: Illustrative output. Observed and predicted length summaries through time.

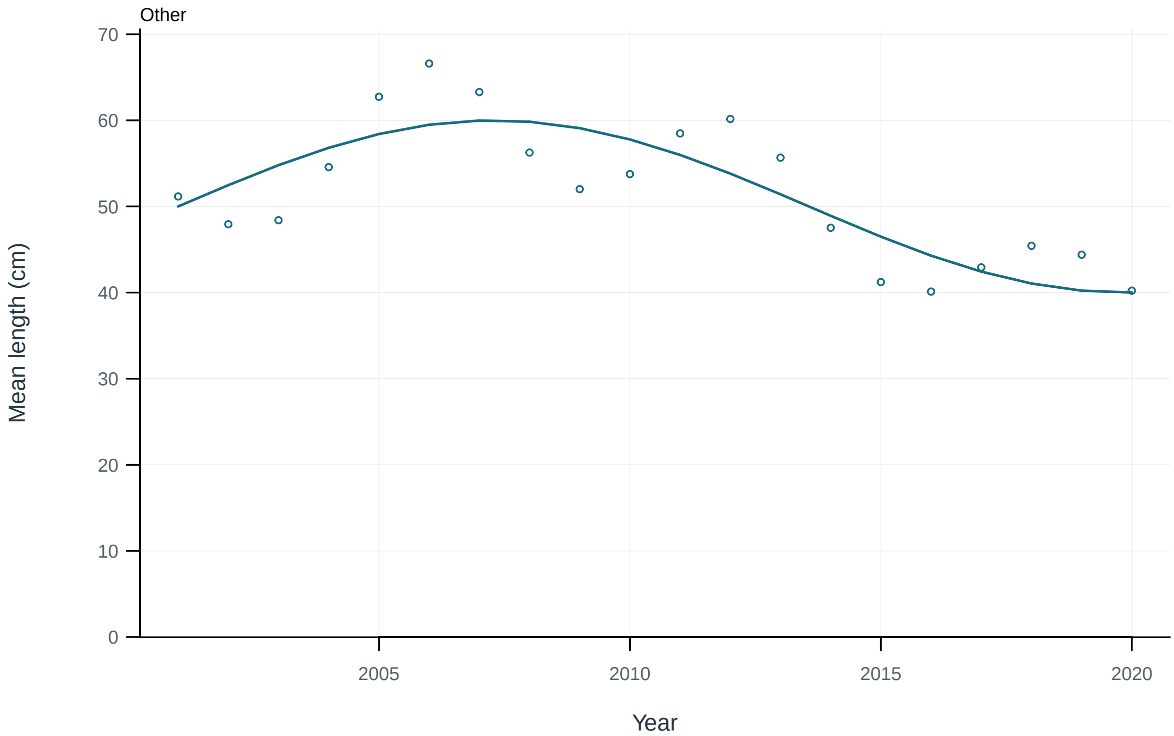

Figure 22: Illustrative output. Observed and predicted length summaries through time.

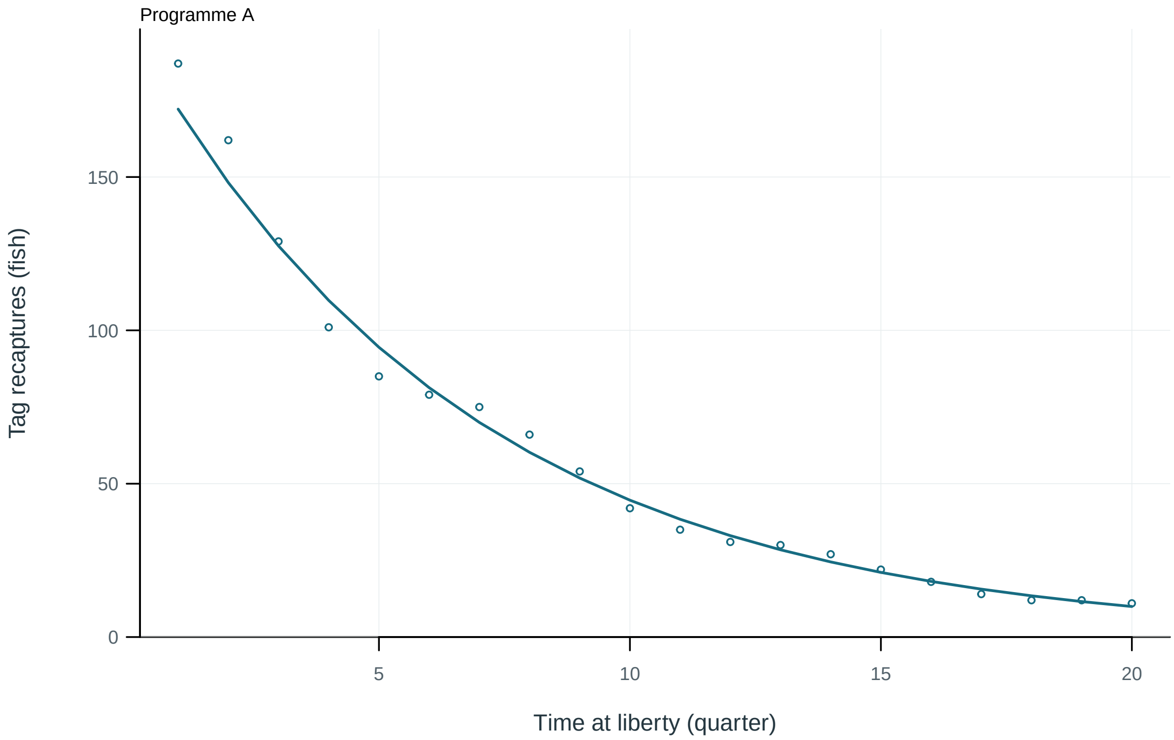

Figure 23: Illustrative output. Observed and predicted tag recaptures by time at liberty.

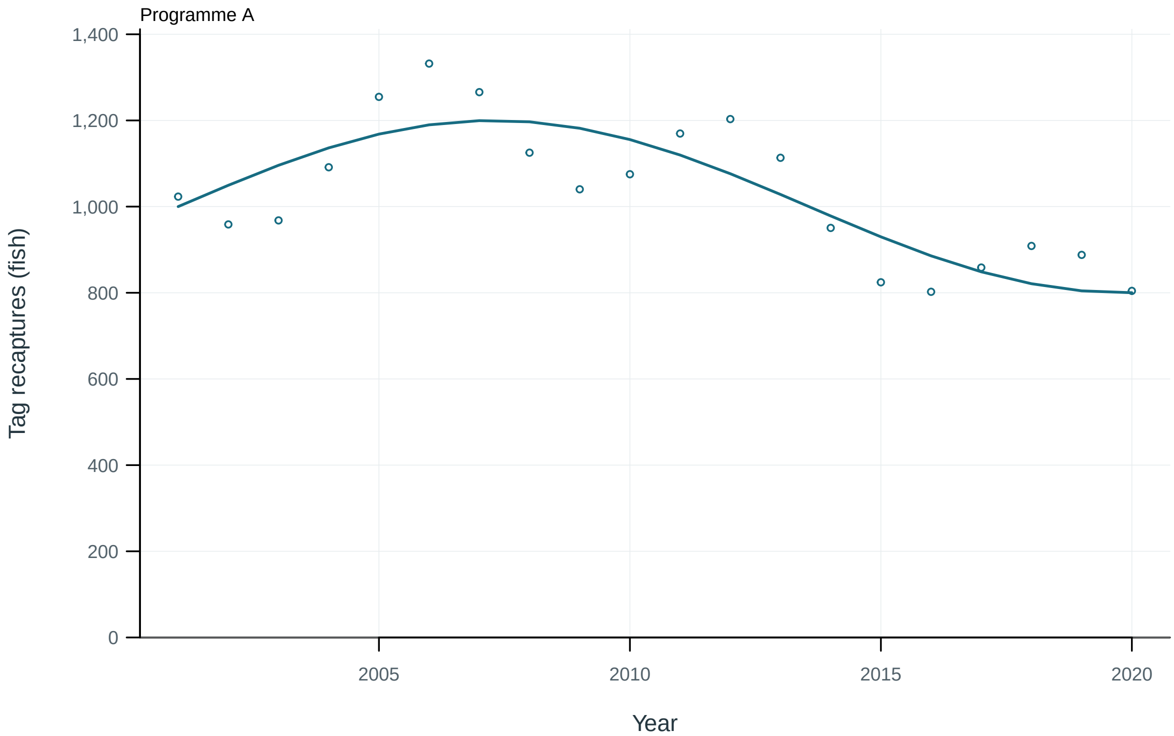

Figure 24: Illustrative output. Observed and predicted tag recaptures through time.

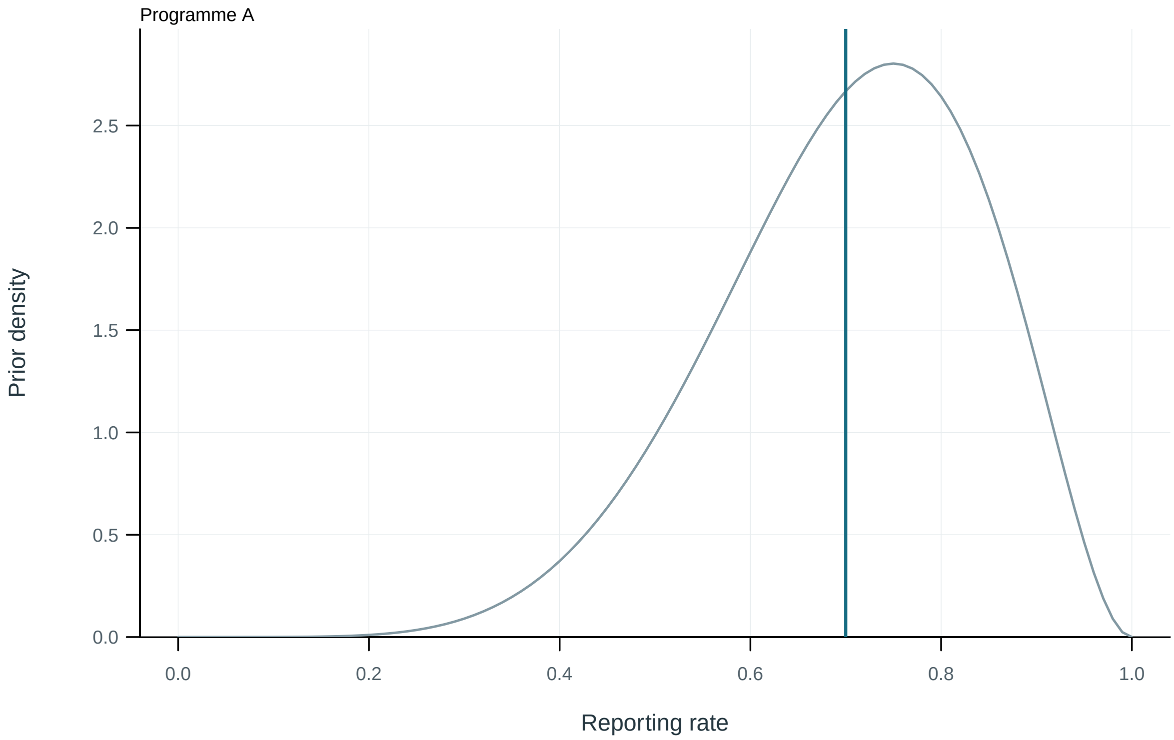

Figure 25: Illustrative output. Estimated tag-reporting rates and supplied priors.

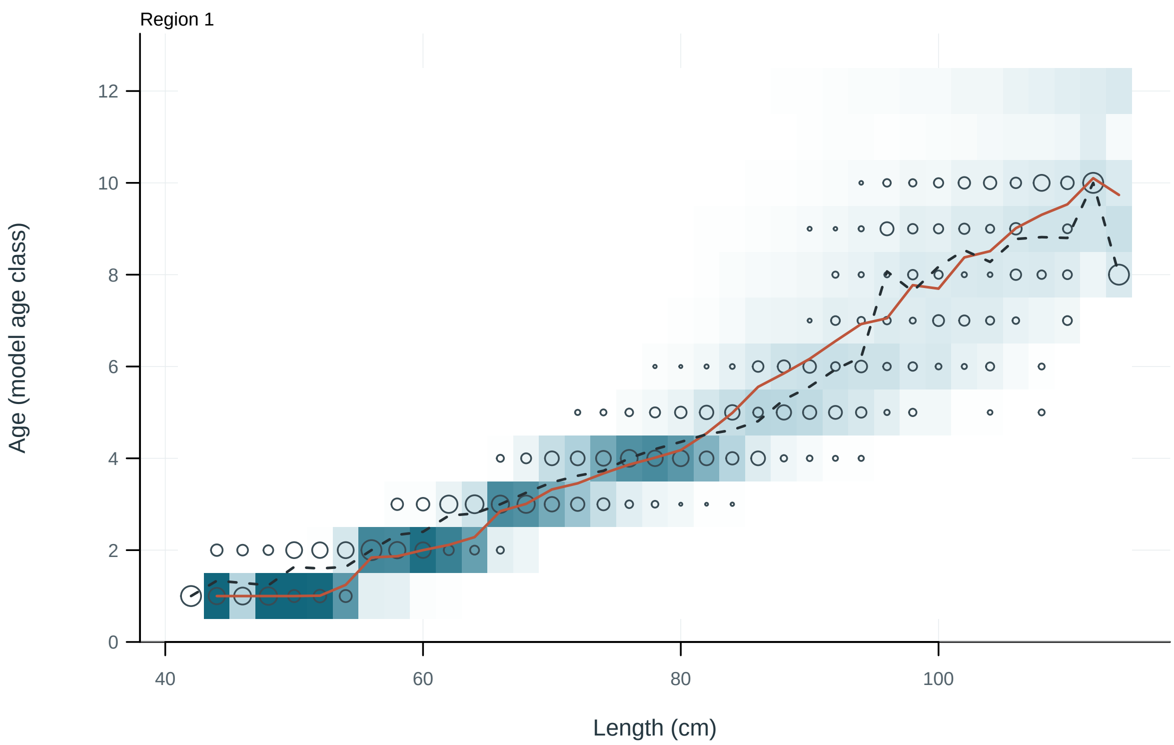

Figure 26: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

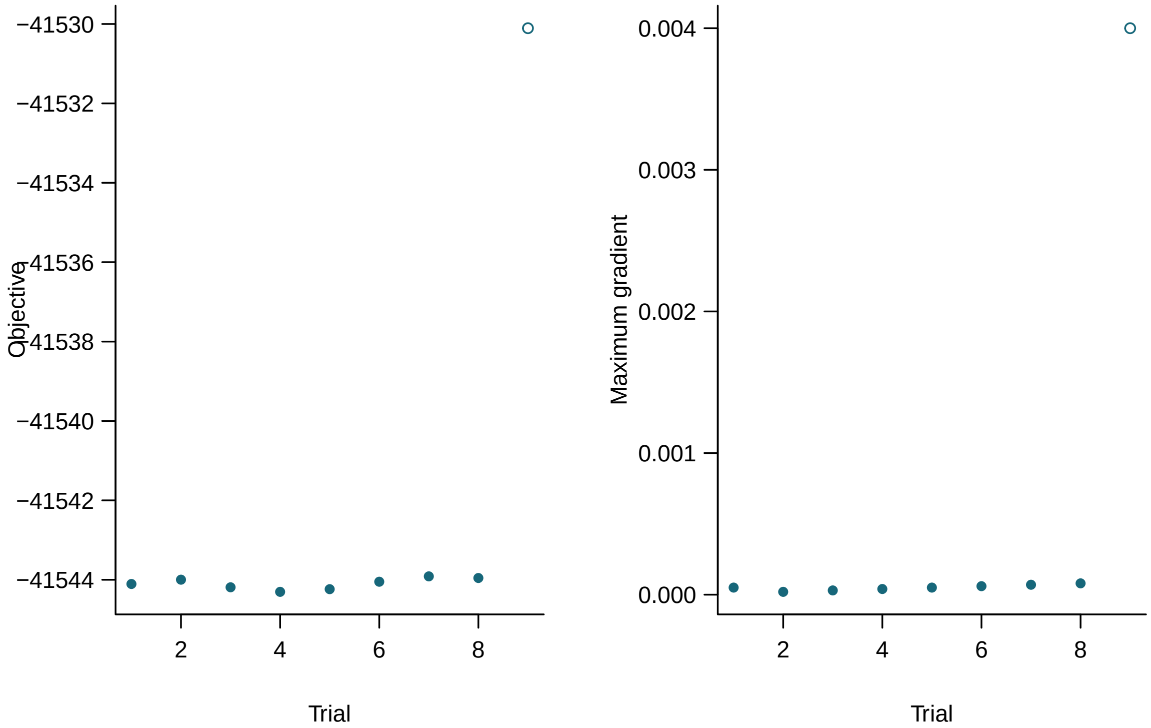

Figure 27: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

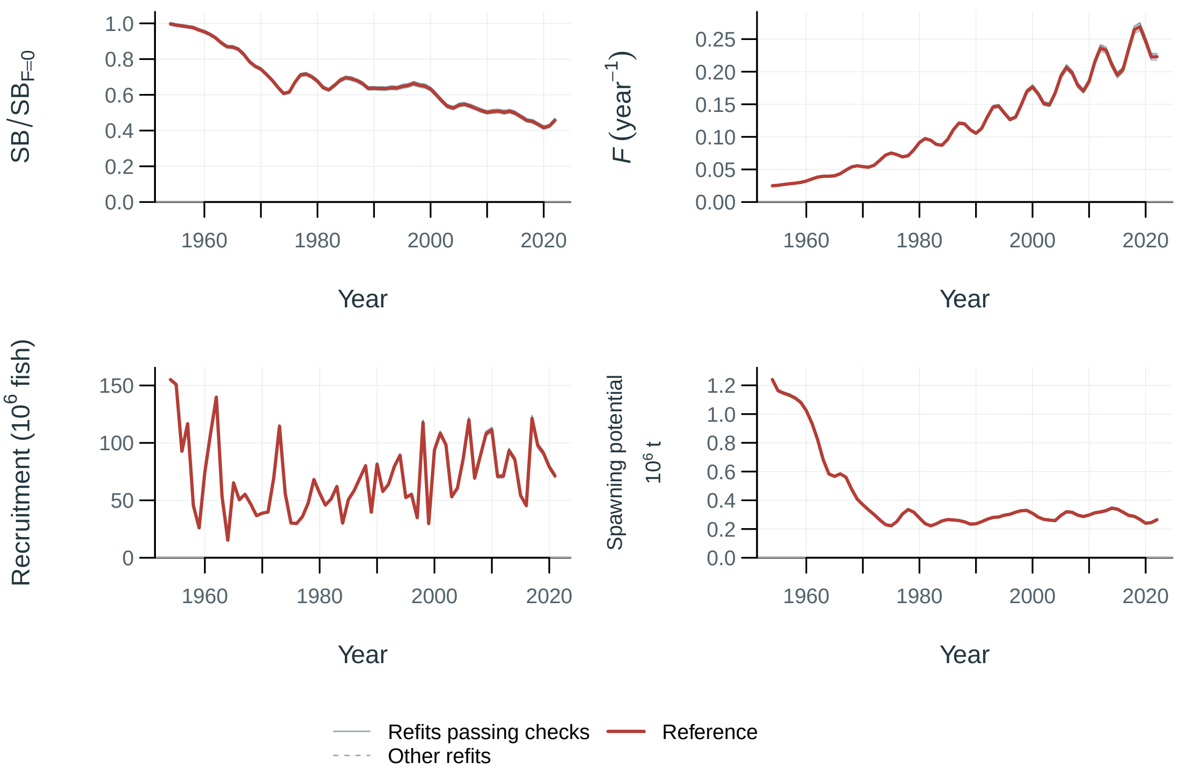

Figure 28: Jitter model trajectories.

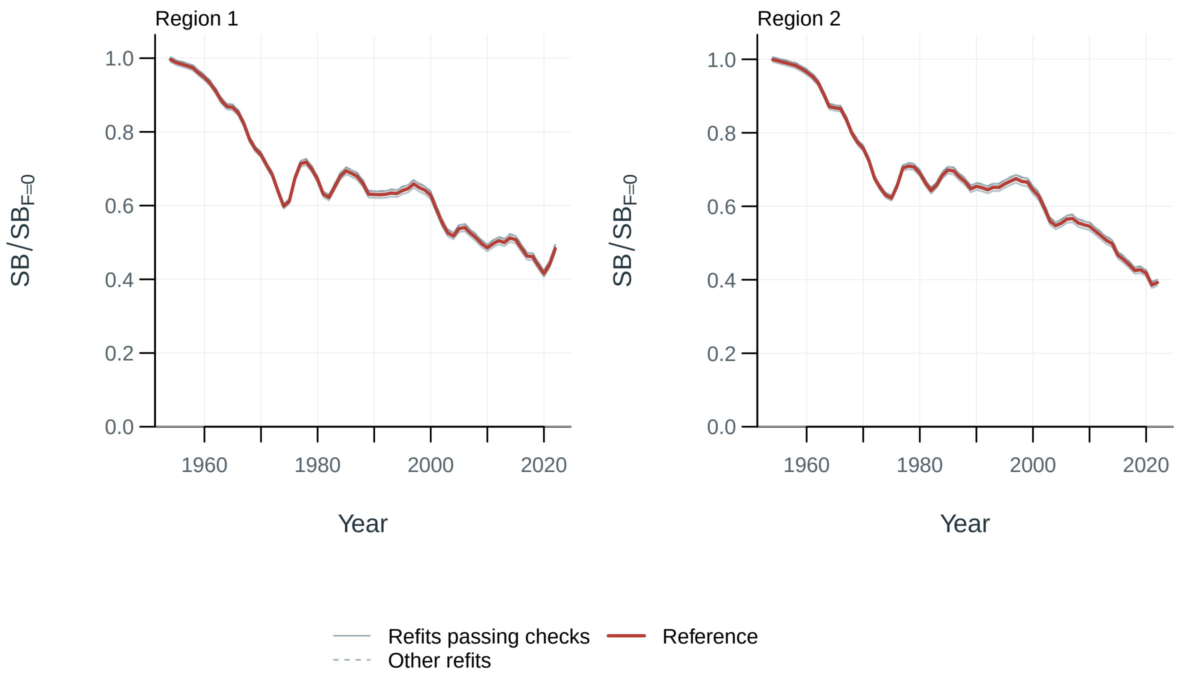

Figure 29: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

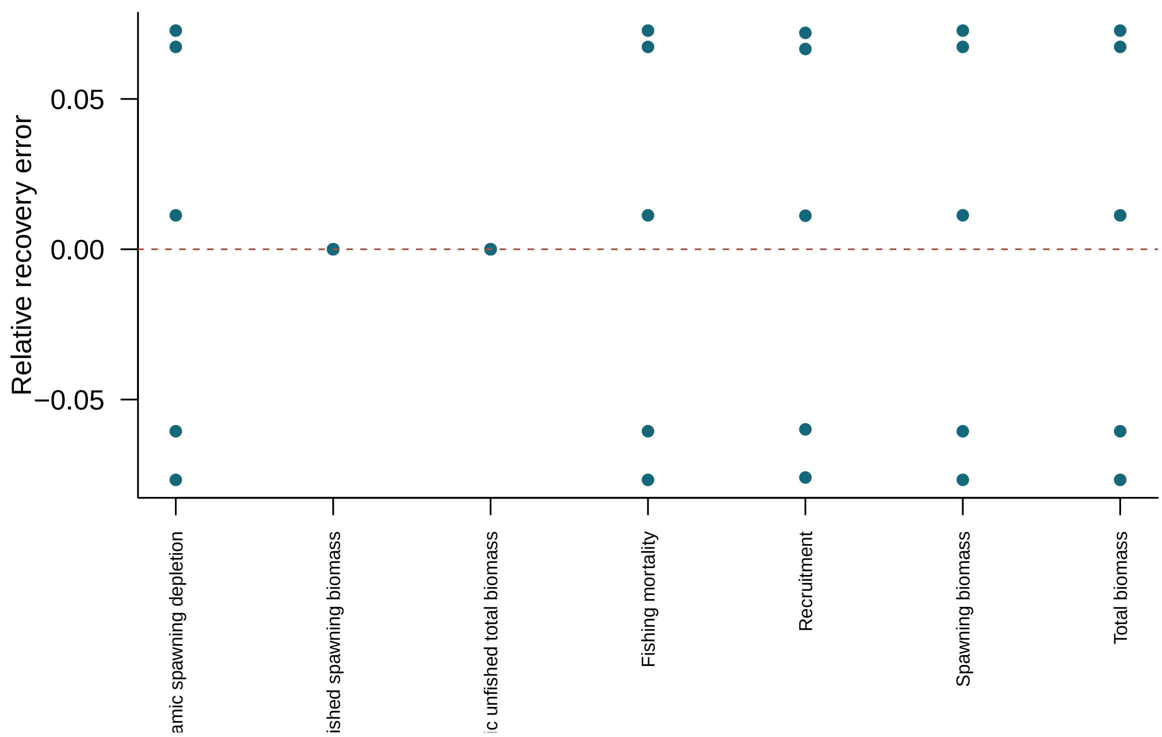

Figure 30: Relative recovery error in terminal quantities for converged simulated-data refits.

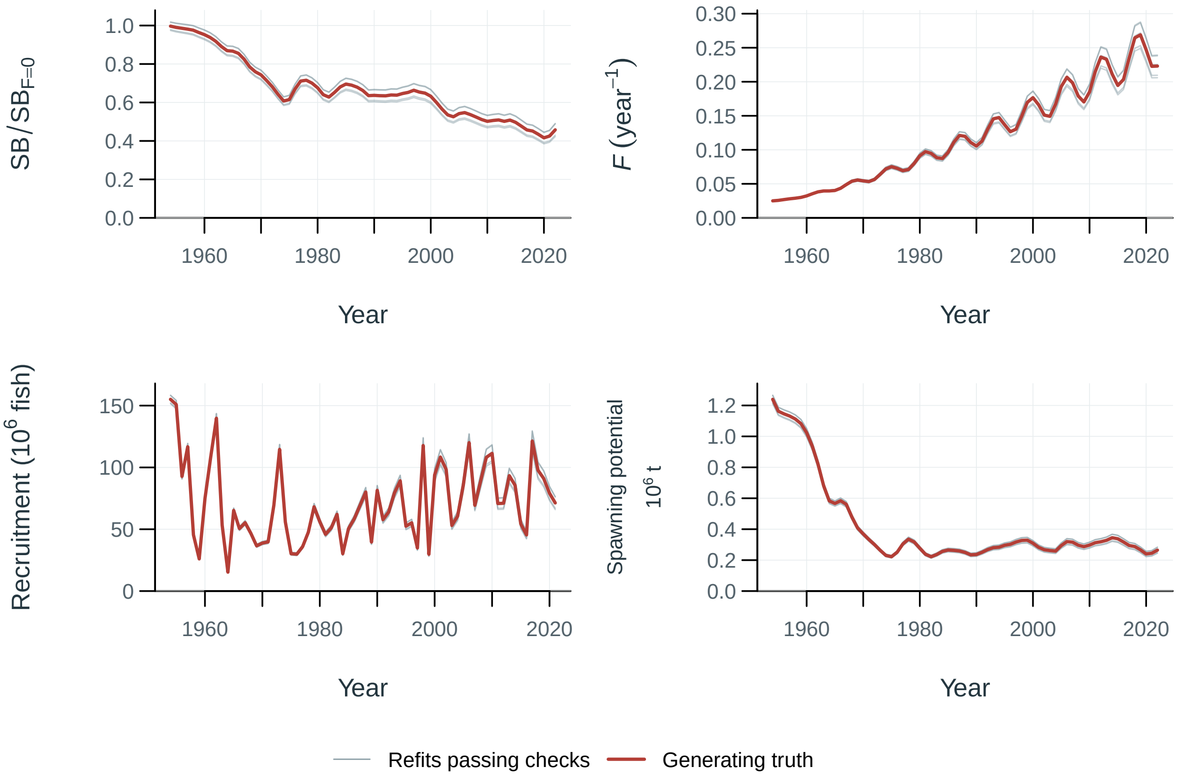

Figure 31: Selftest model trajectories.

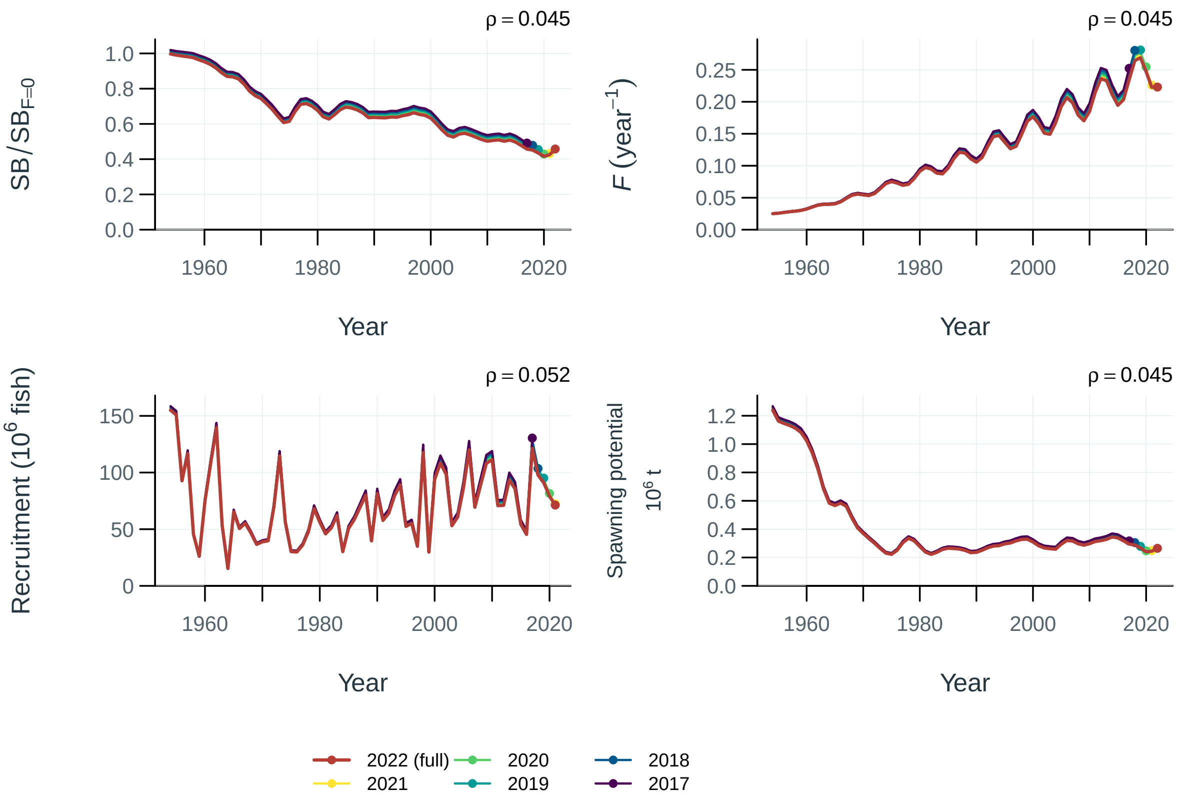

Figure 32: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

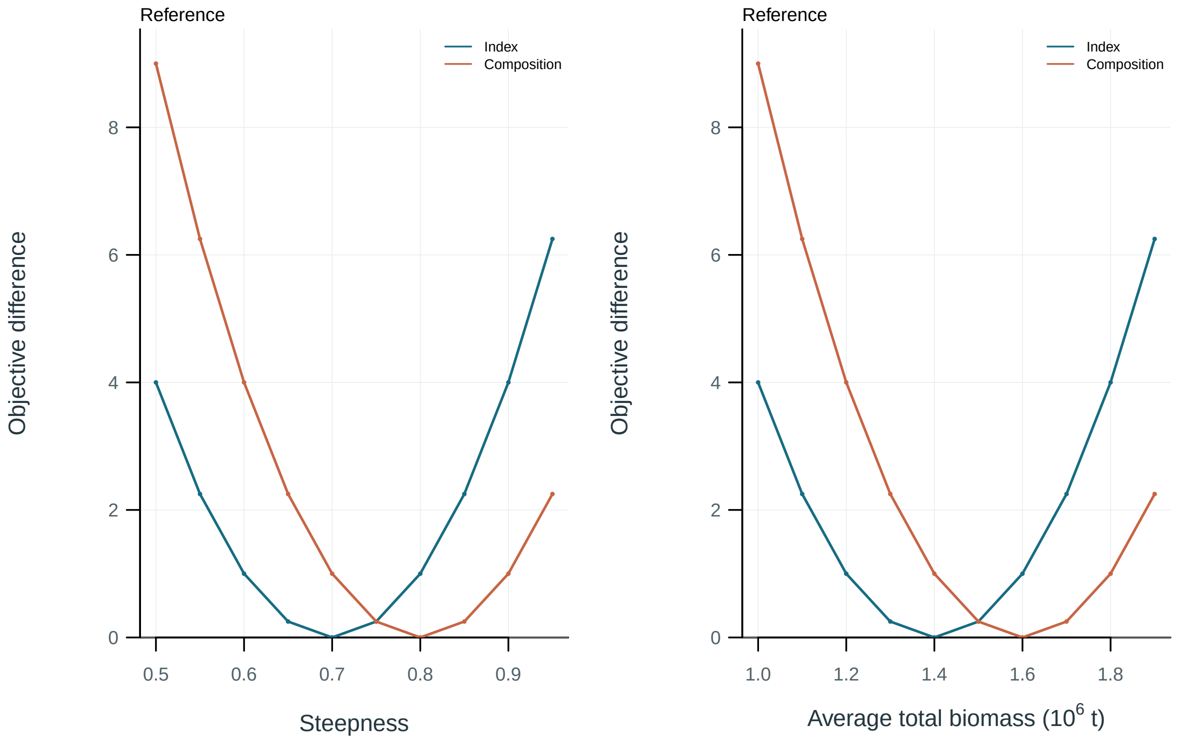

Figure 33: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

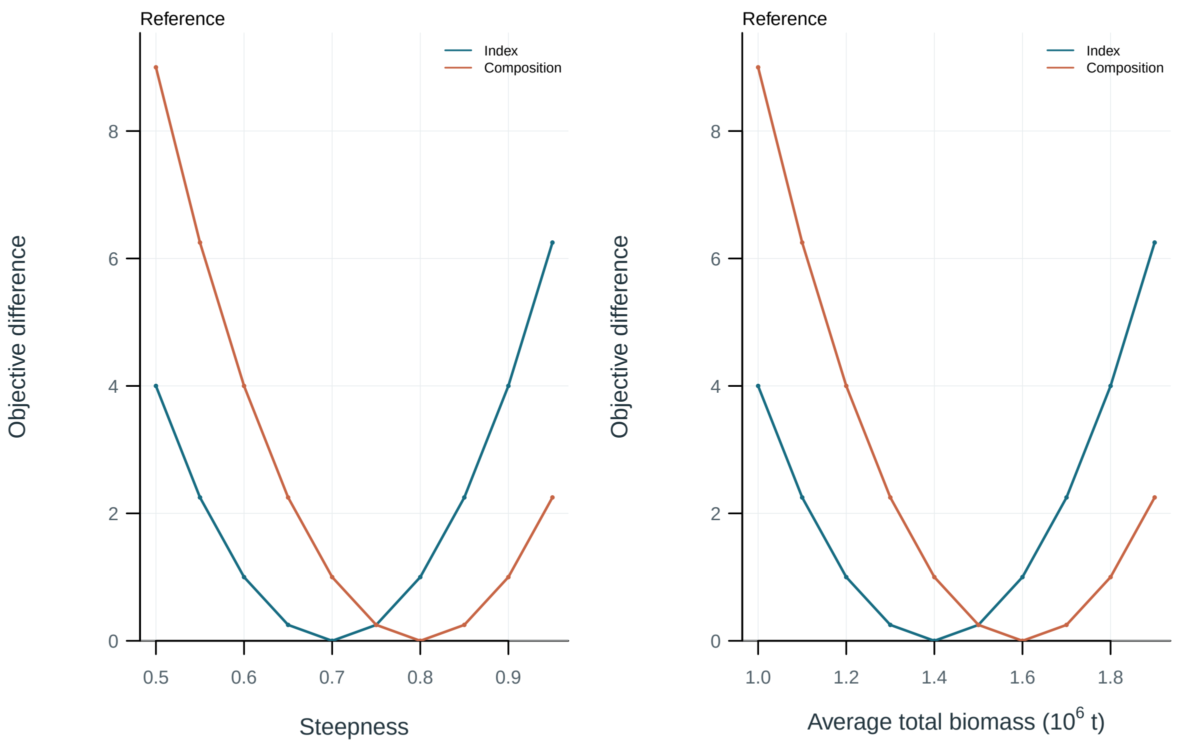

Figure 34: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

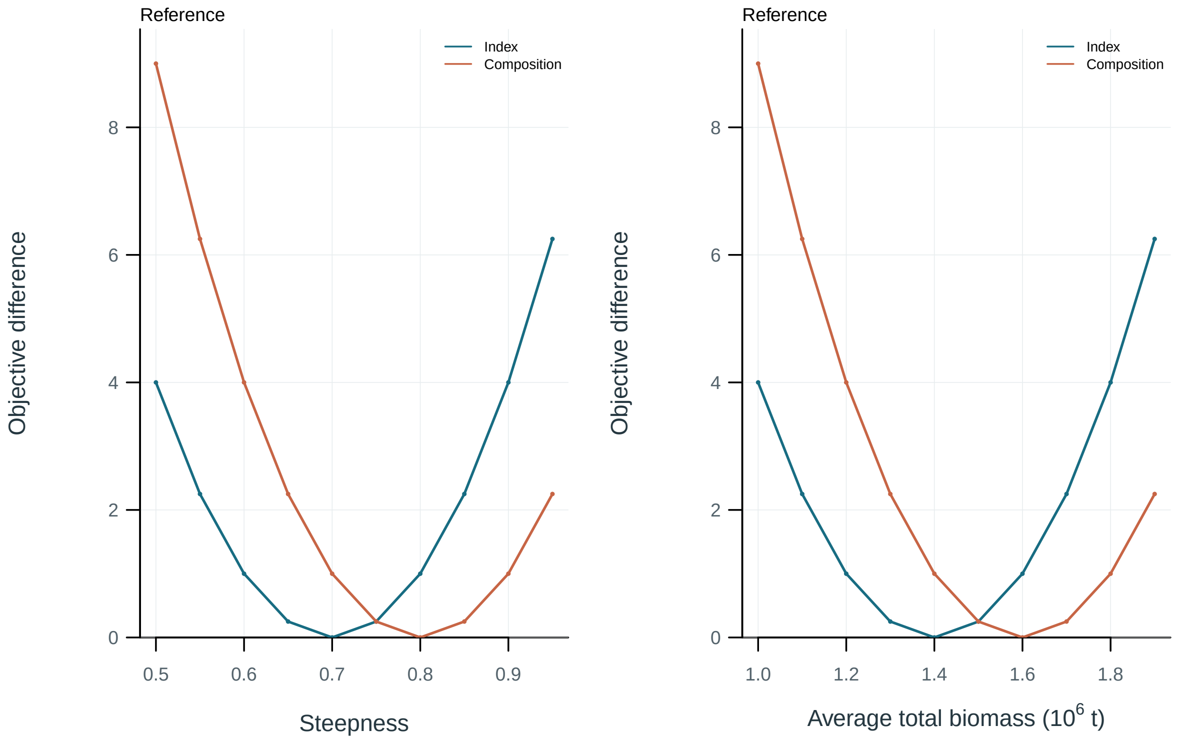

Figure 35: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

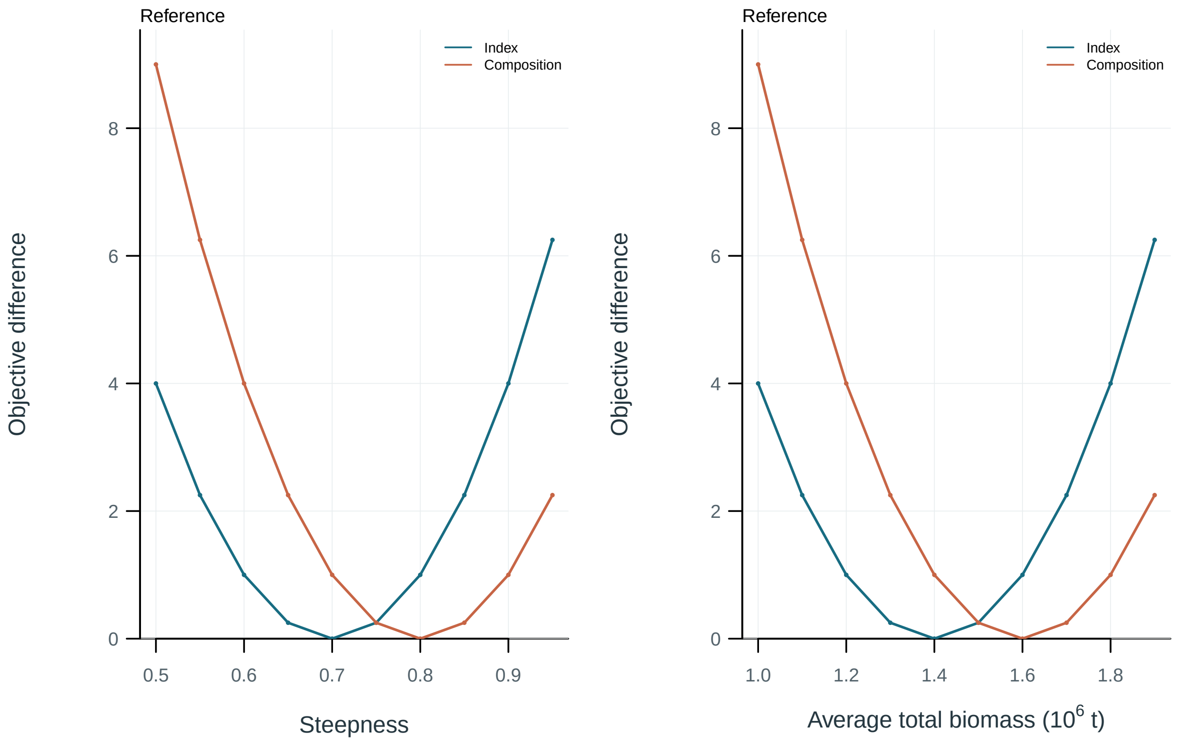

Figure 36: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

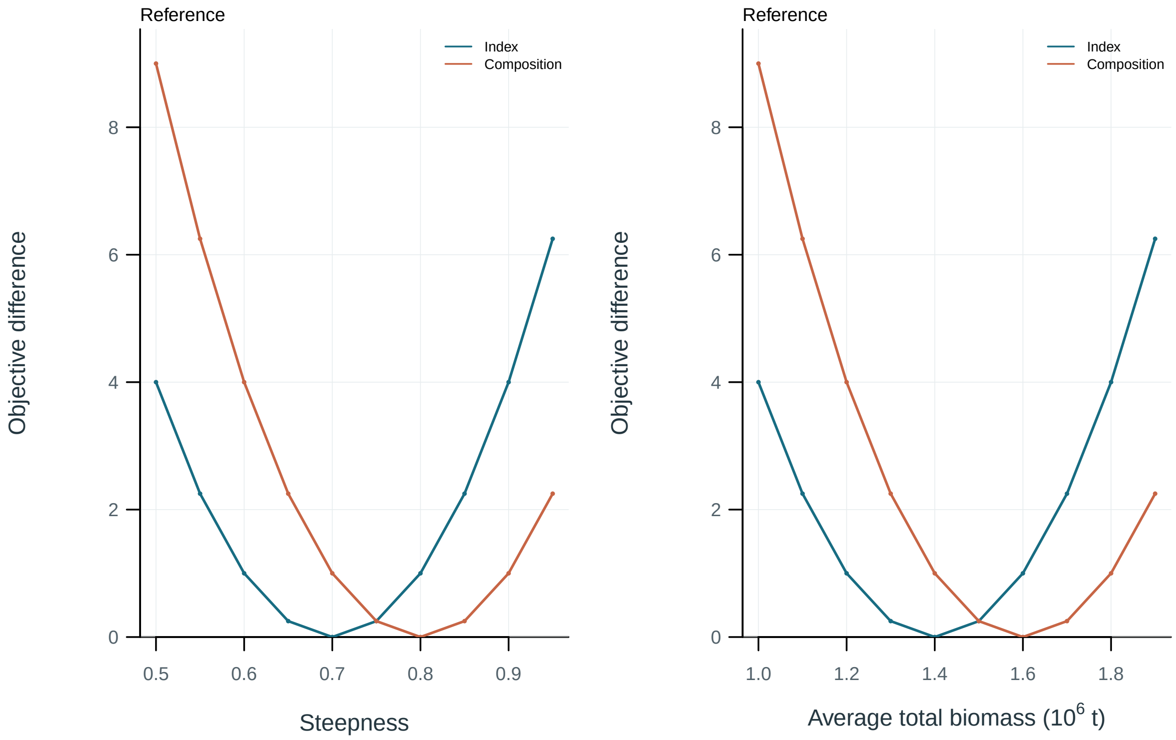

Figure 37: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

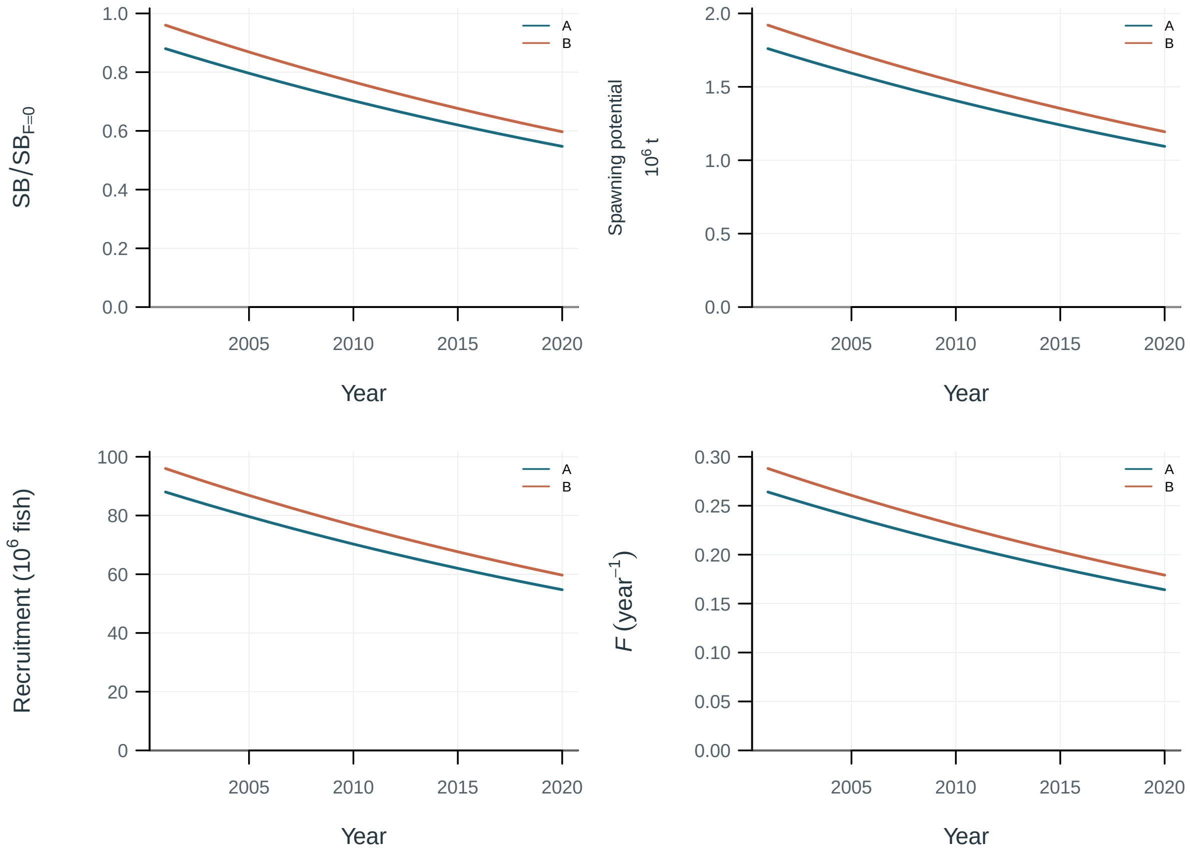

Figure 38: Illustrative output. Diagnostic and production-model trajectories.

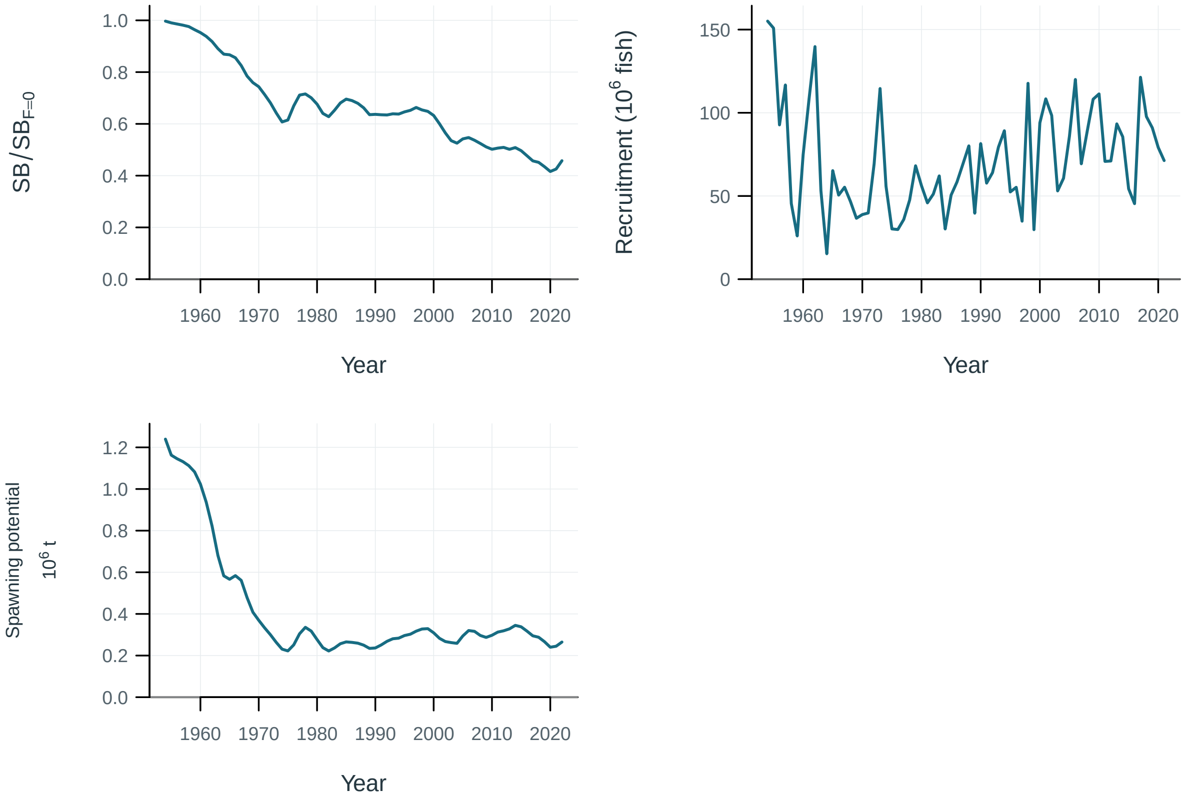

Figure 39: Estimated population trajectories.

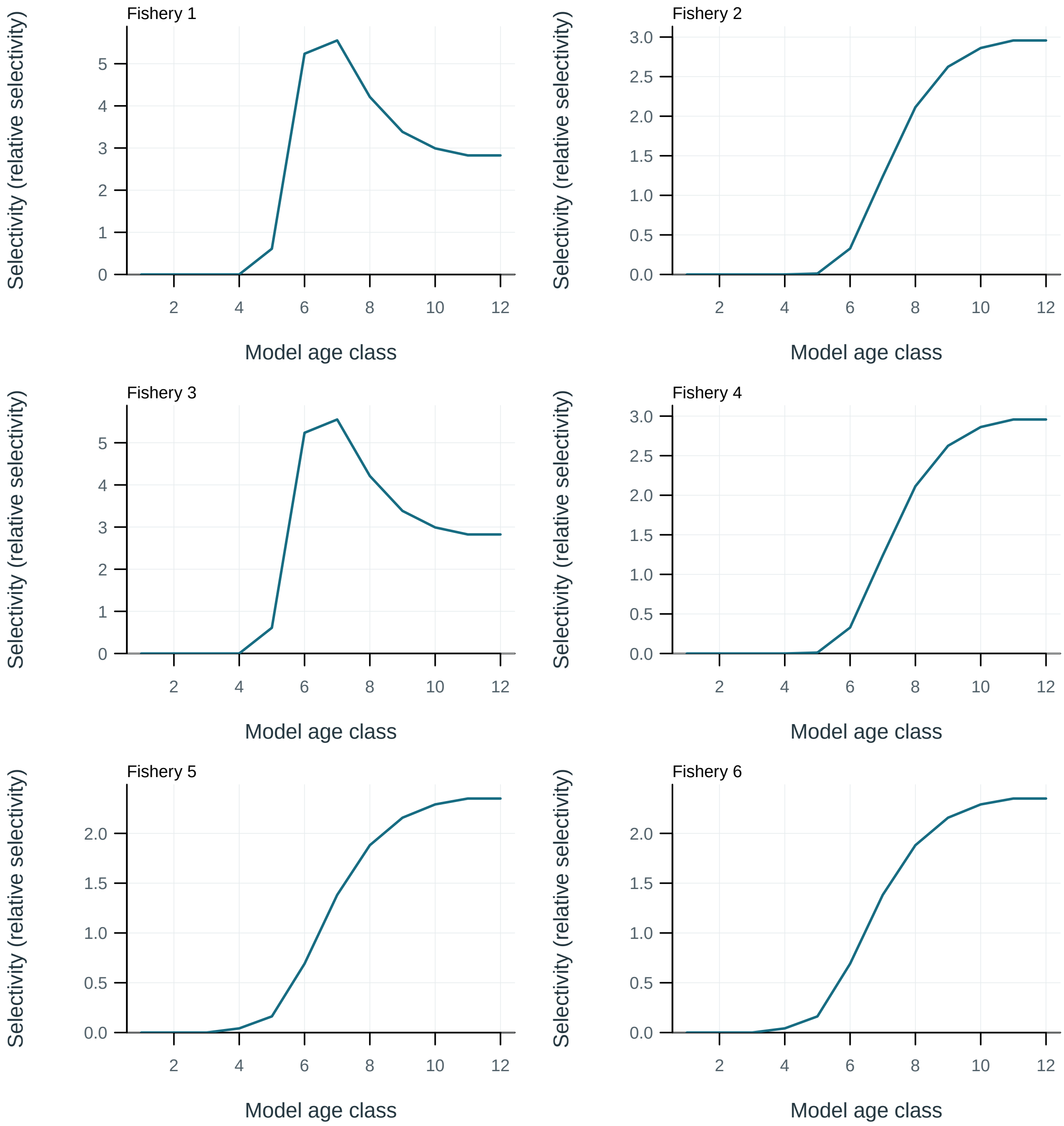

Figure 40: Fishery selectivity at age.

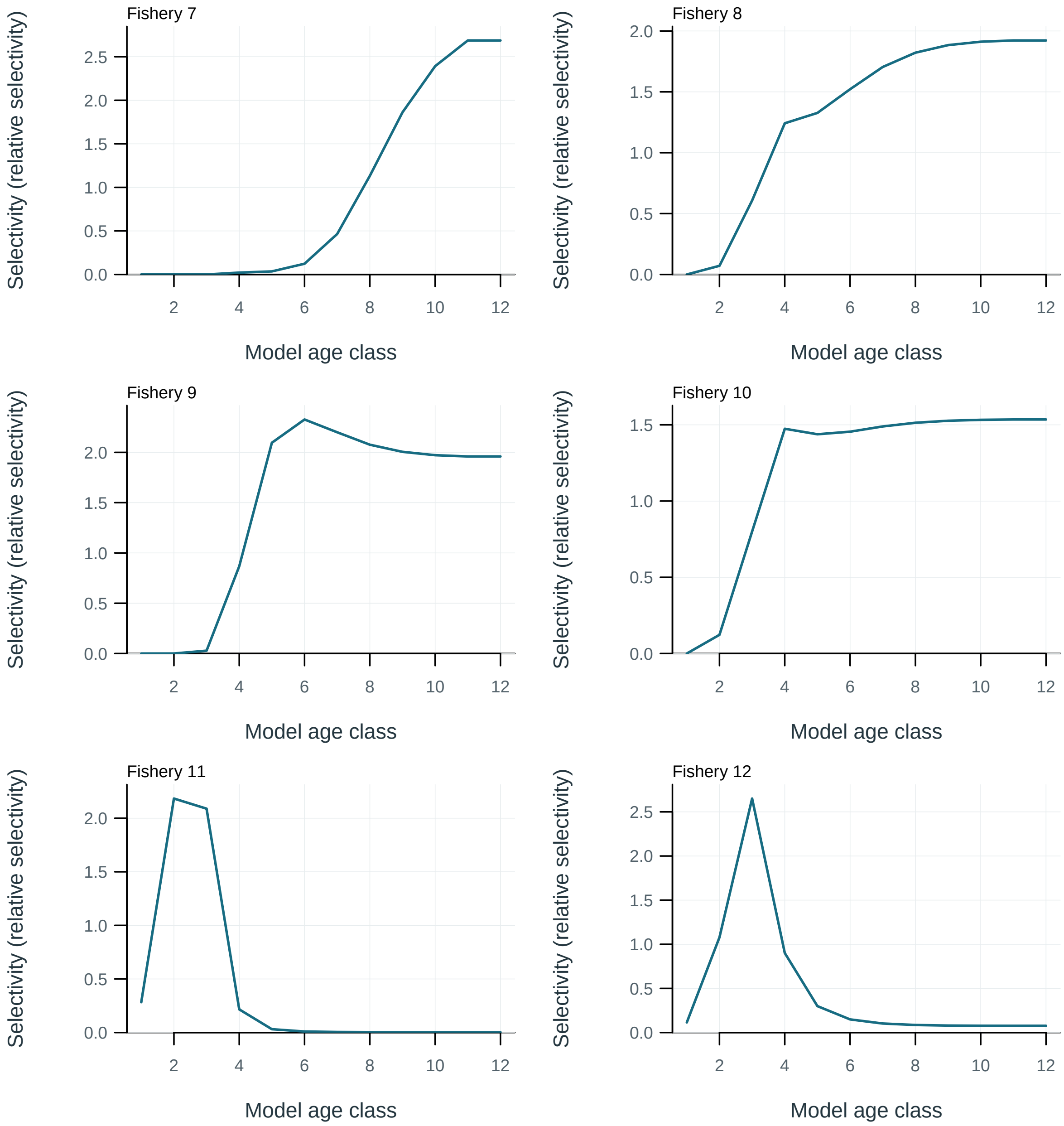

Figure 41: Fishery selectivity at age.

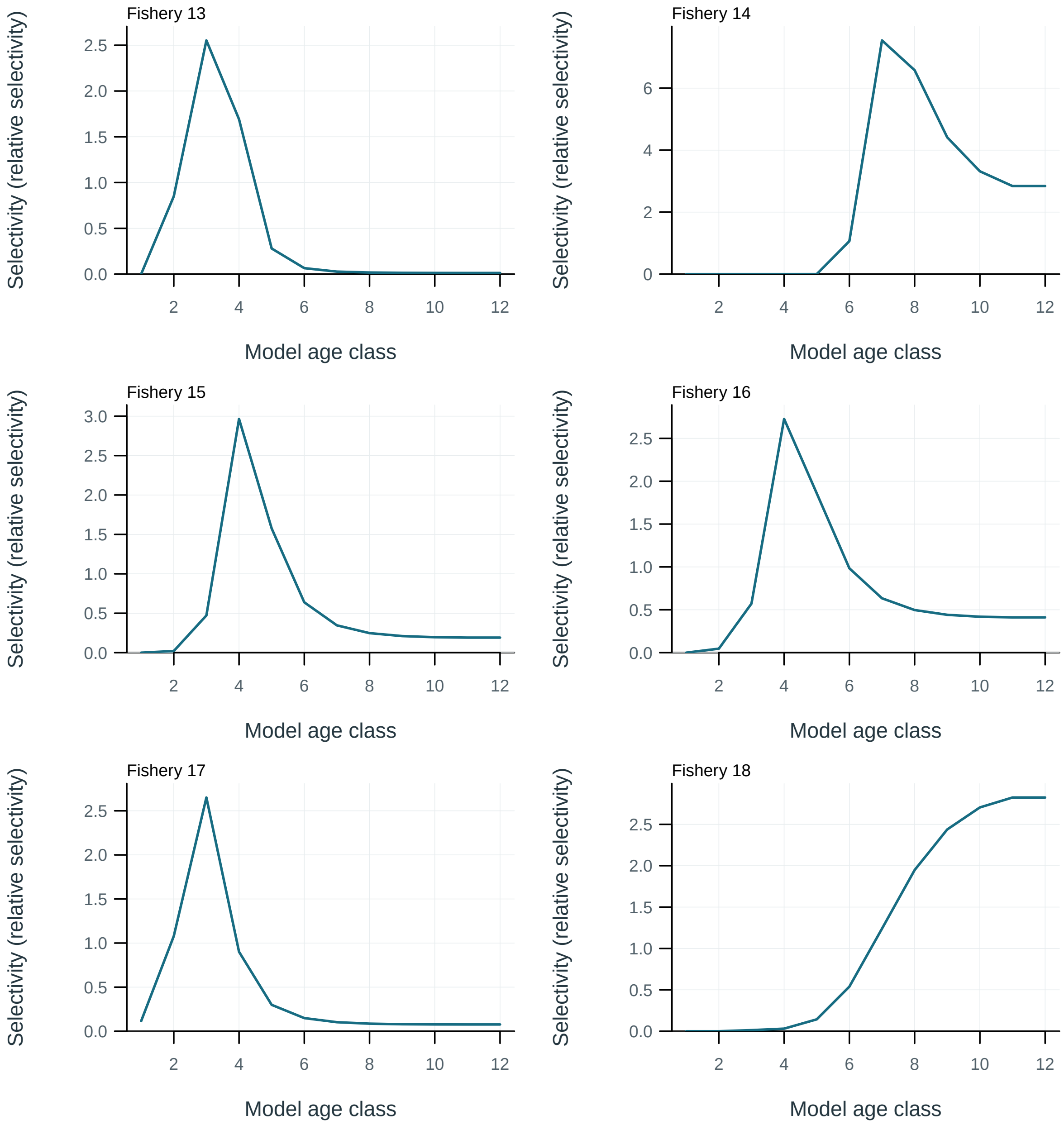

Figure 42: Fishery selectivity at age.

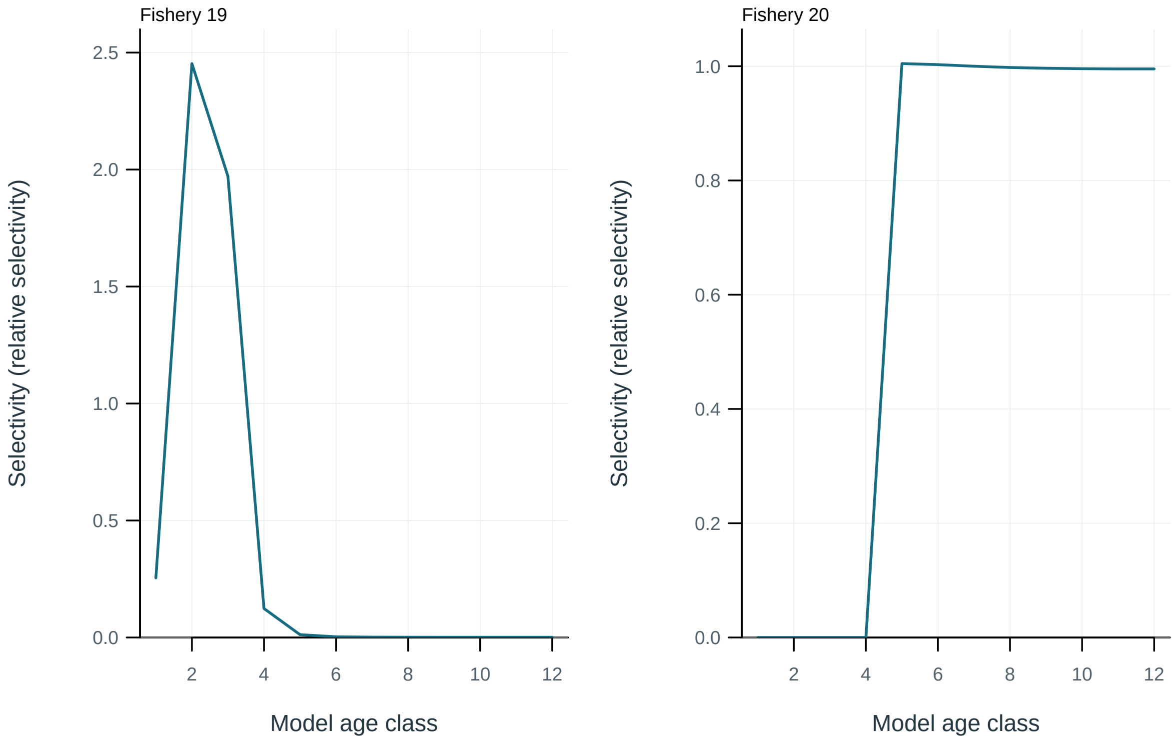

Figure 43: Fishery selectivity at age.

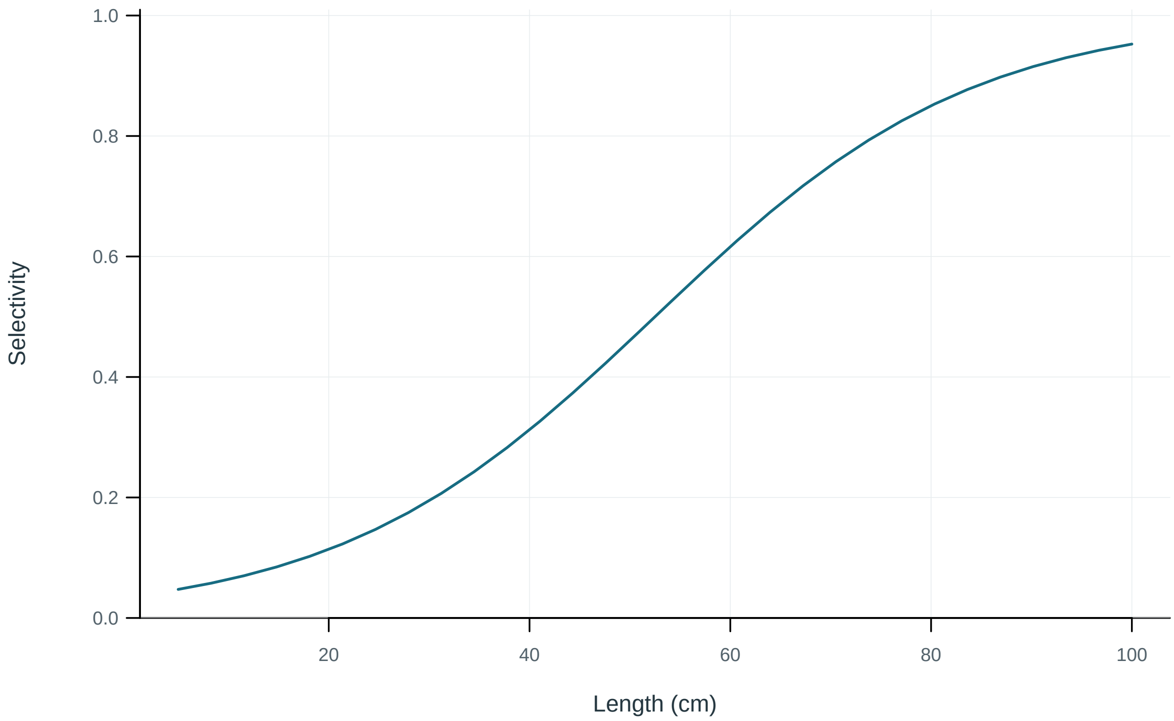

Figure 44: Illustrative output. Selectivity at length by fishery.

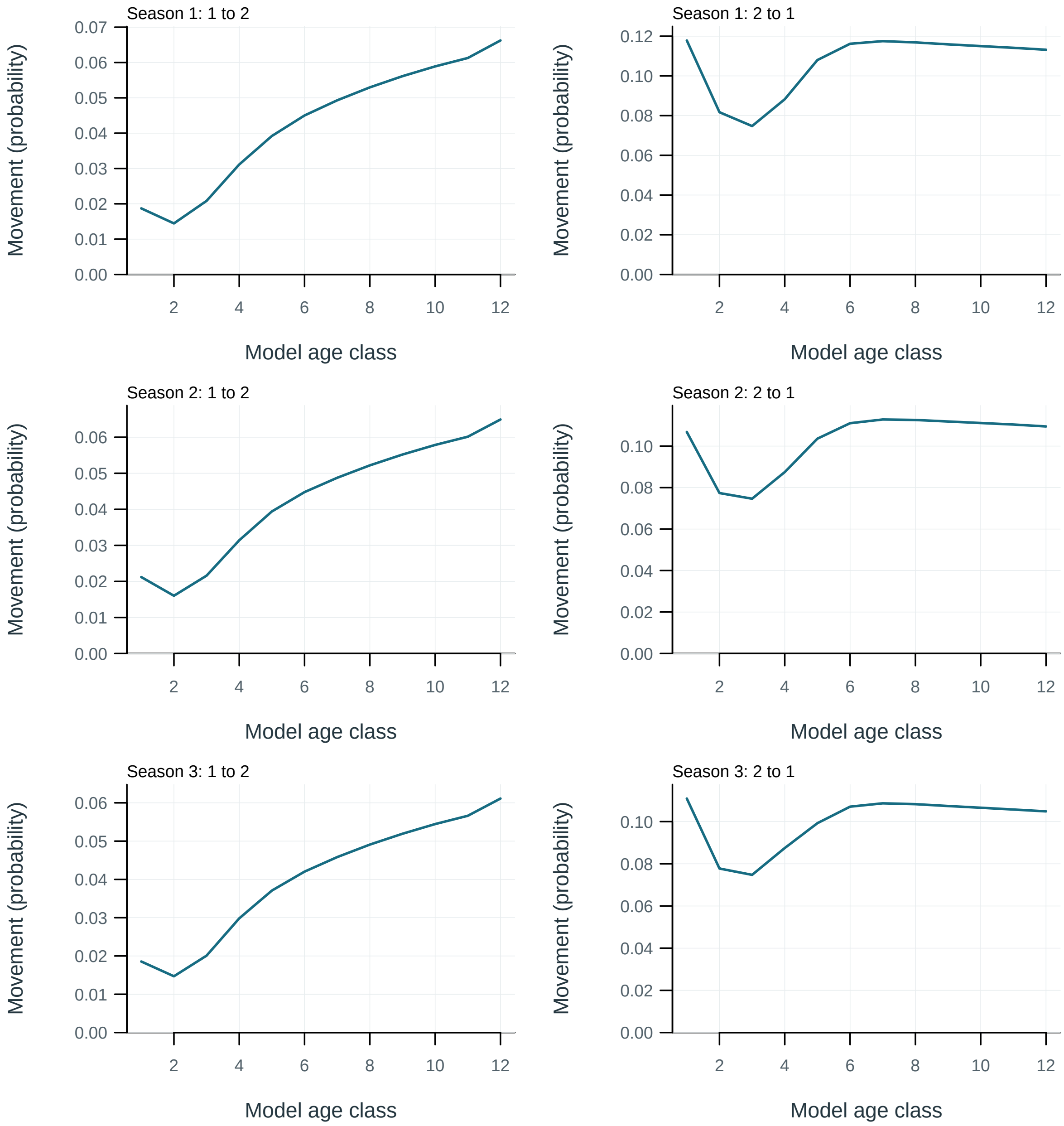

Figure 45: Probability of movement from source to destination region per movement event.

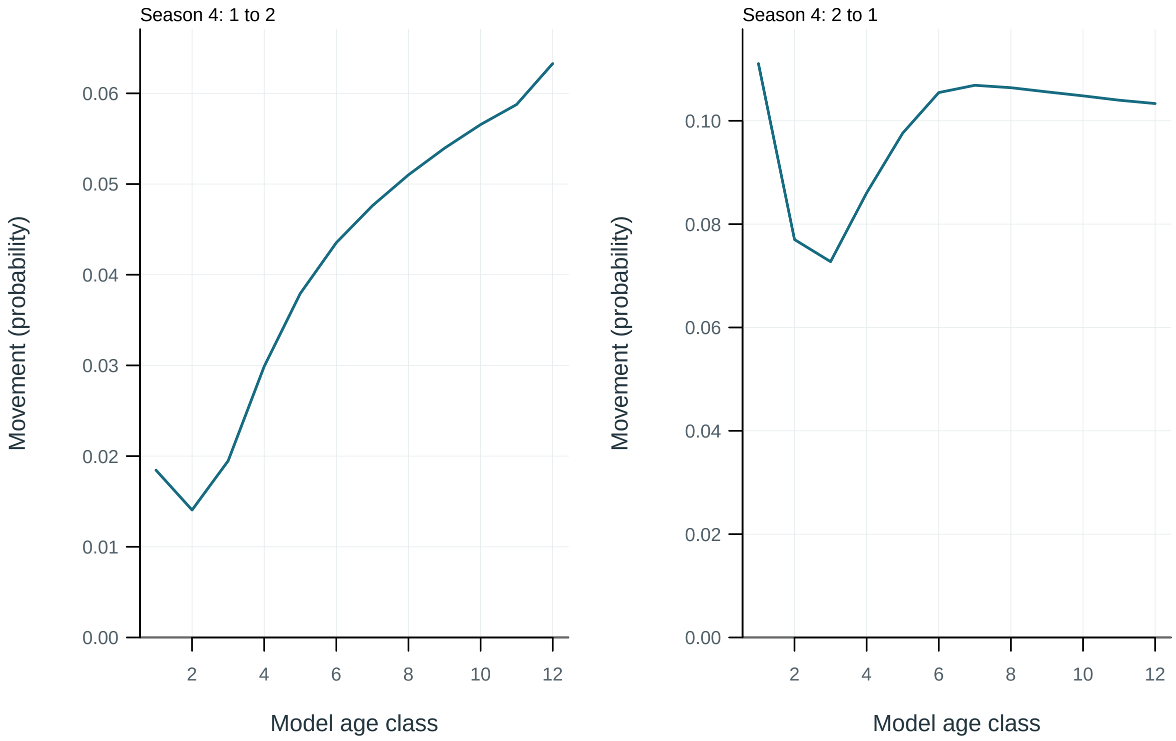

Figure 46: Probability of movement from source to destination region per movement event.

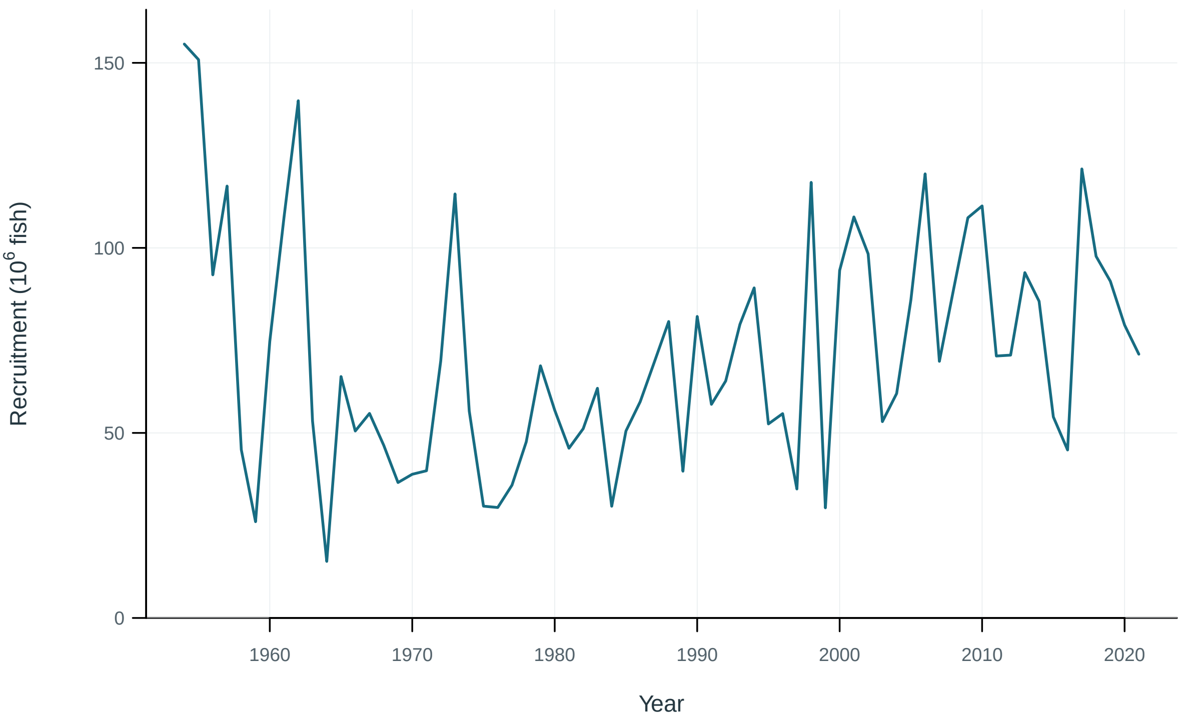

Figure 47: Recruitment by model region.

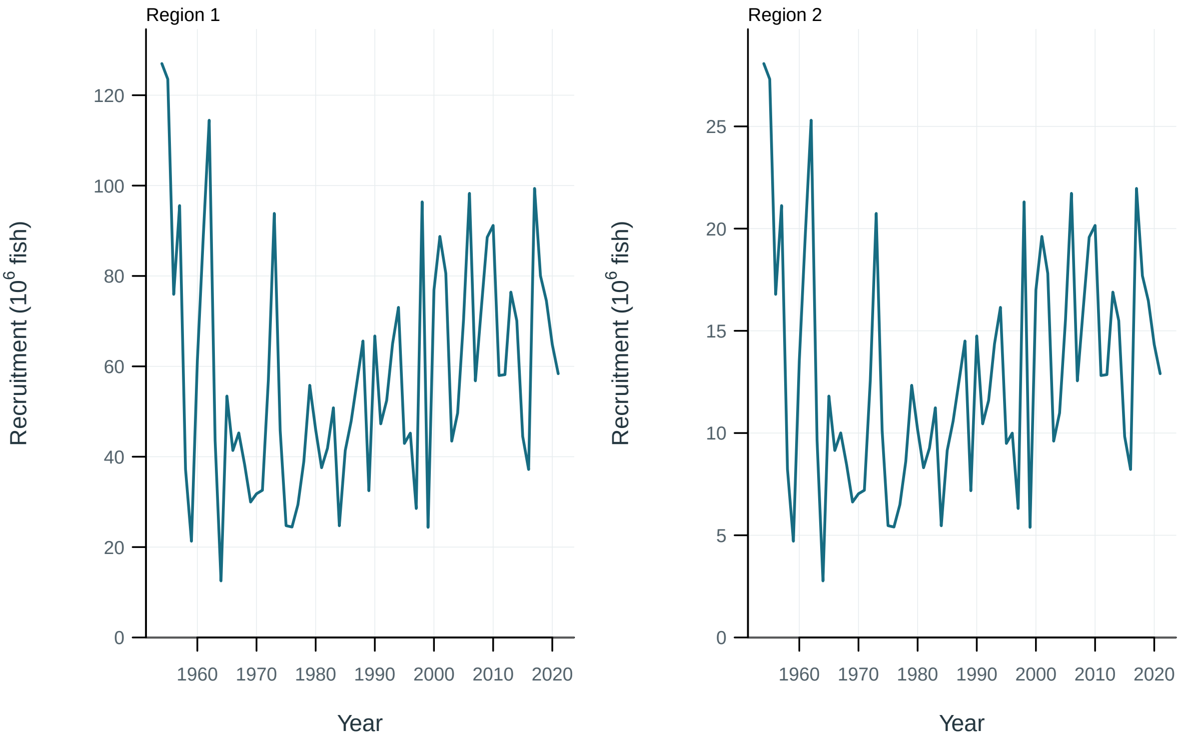

Figure 48: Recruitment by model region.

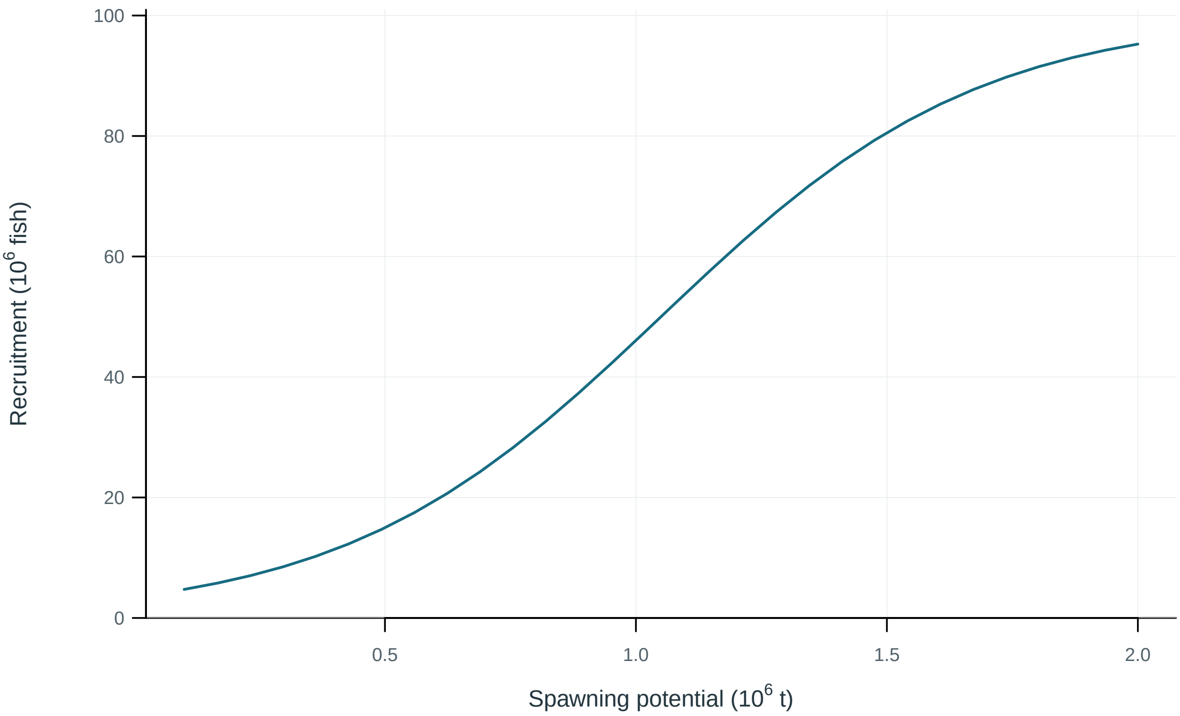

Figure 49: Illustrative output. Stock–recruitment observations and fitted relationship.

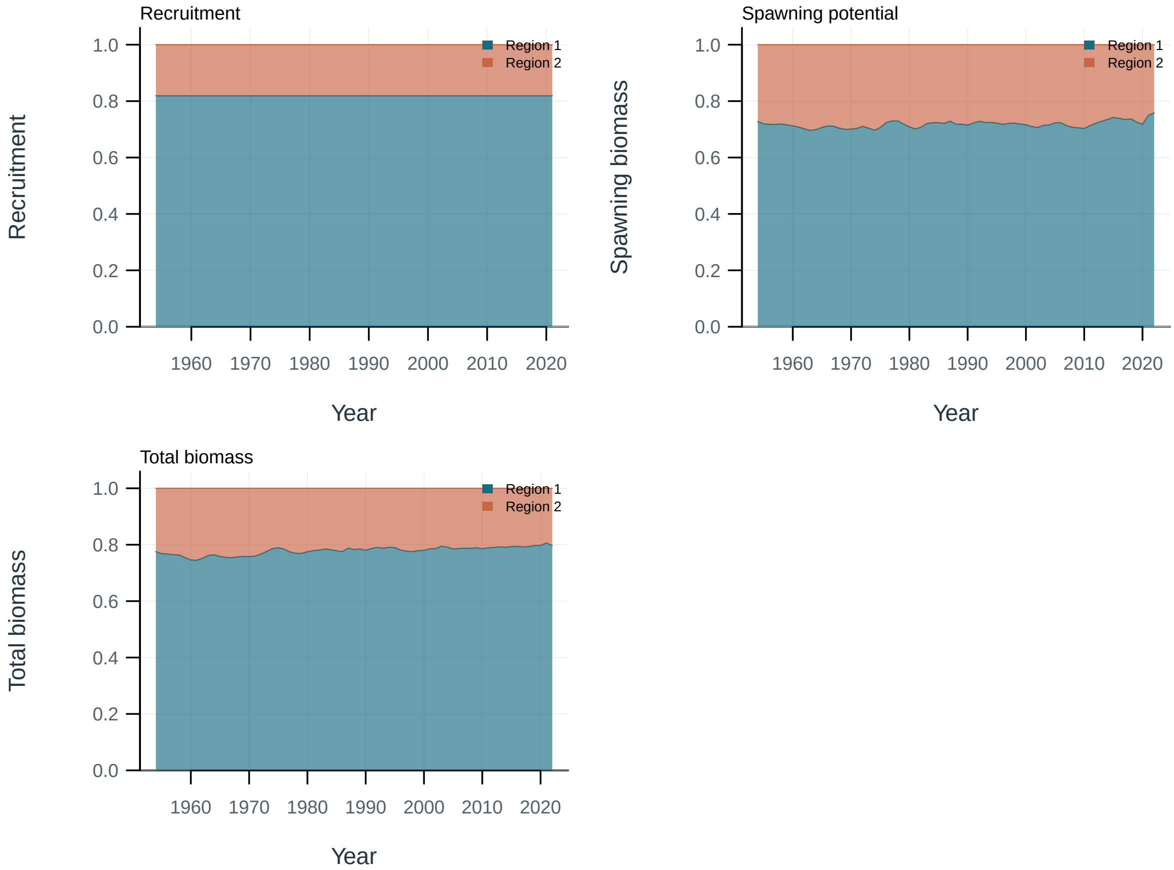

Figure 50: Regional shares of population quantities.

Figure 51: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

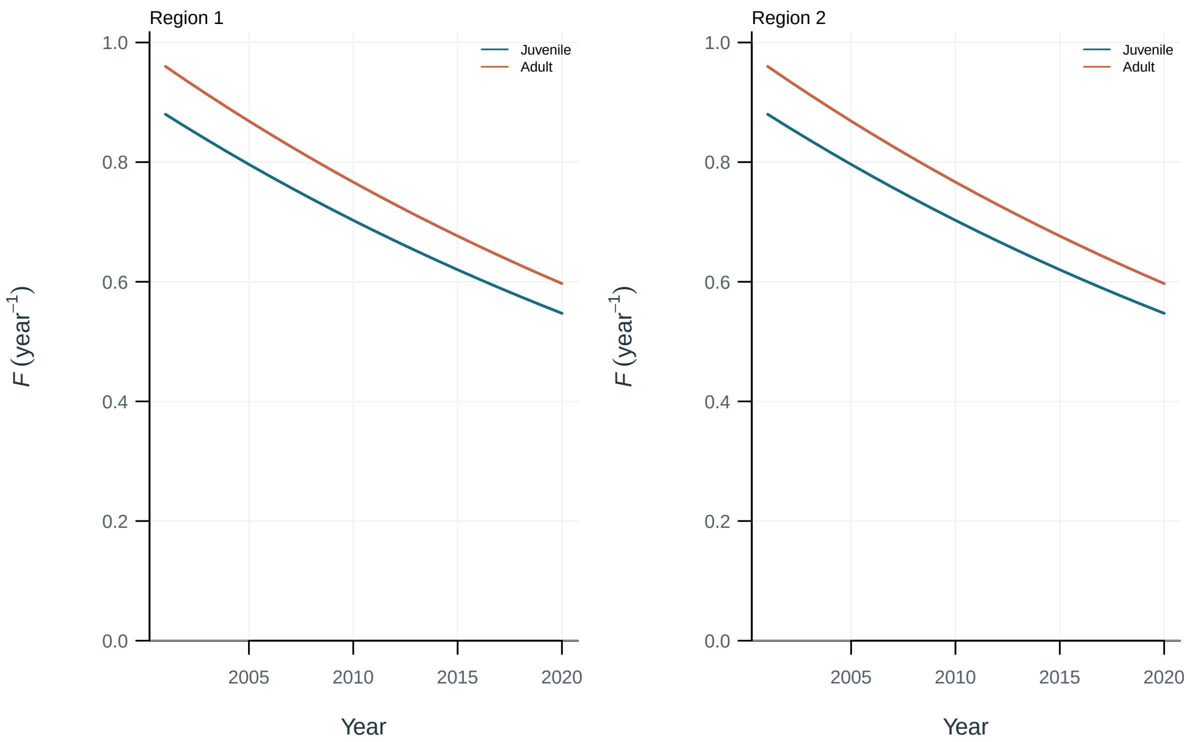

Figure 52: Illustrative output. Fishing mortality by life stage and region.

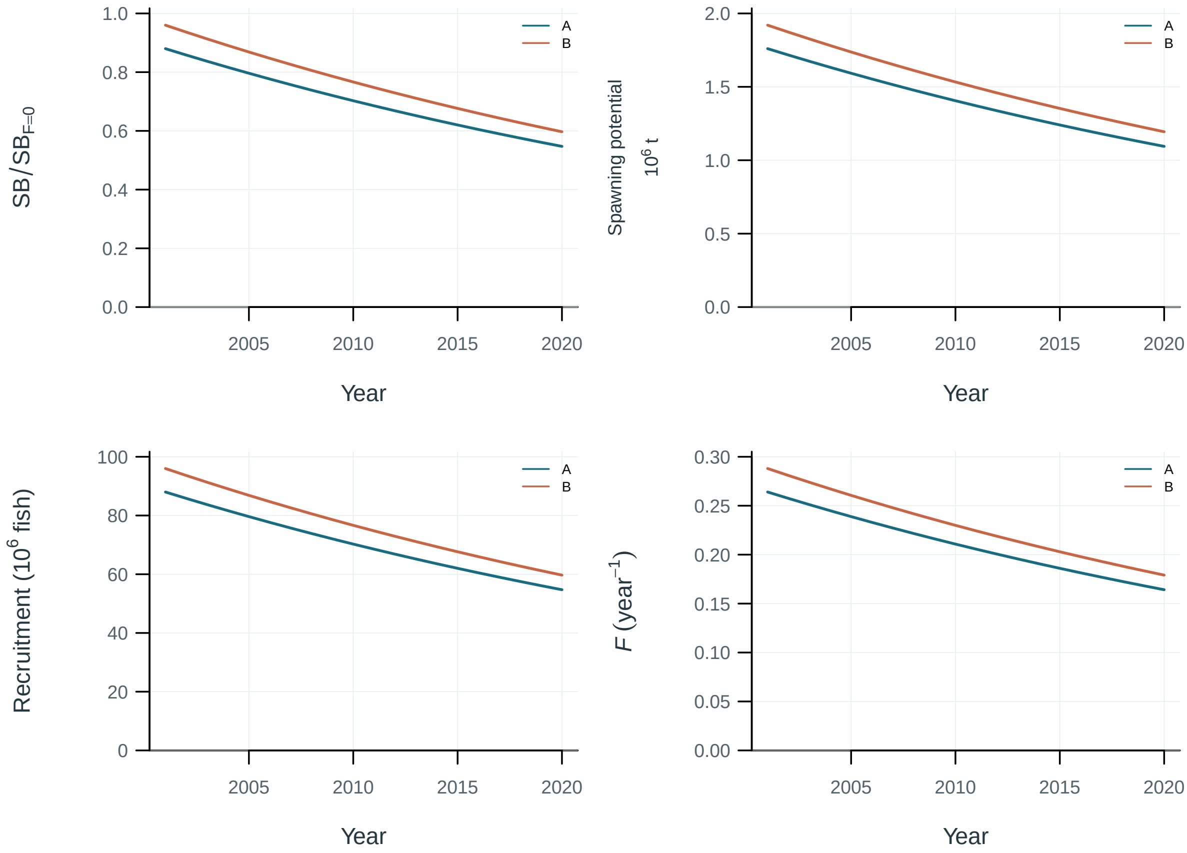

Figure 53: Population trajectories across the stated sensitivity models.

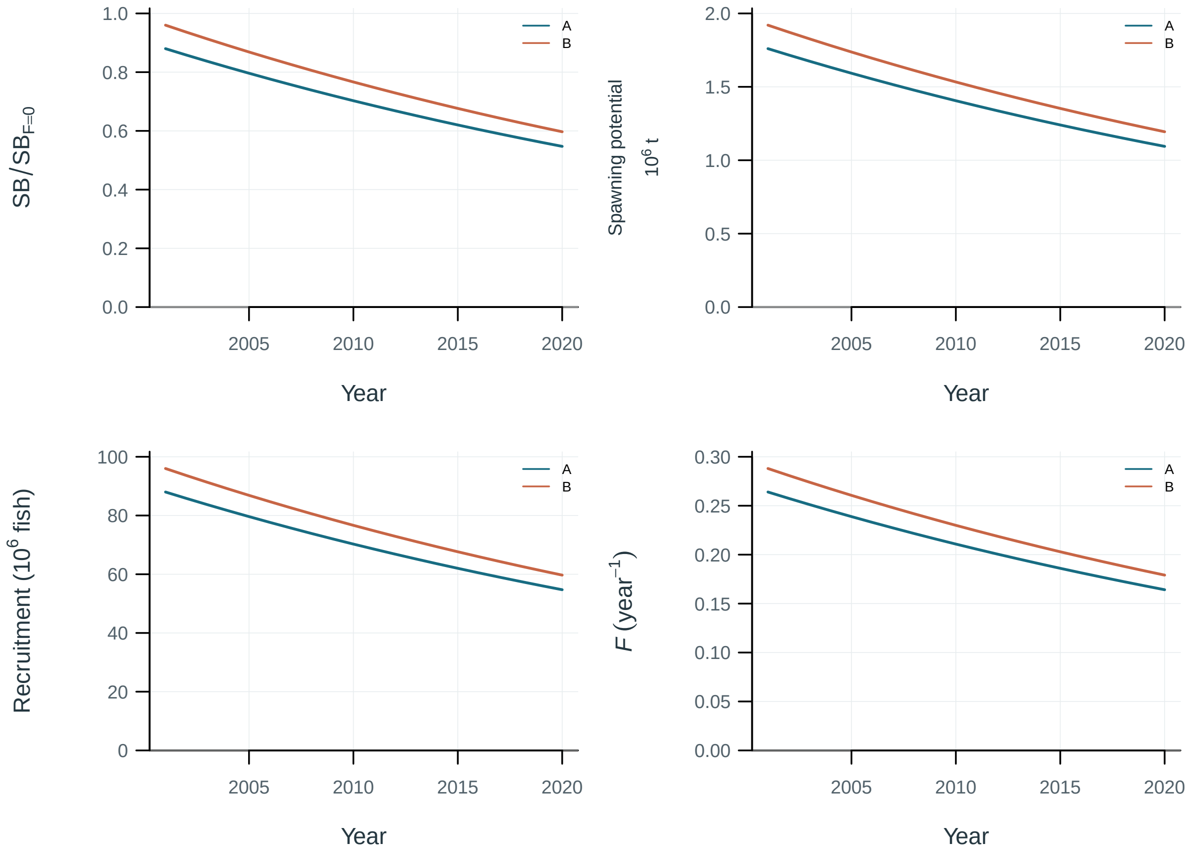

Figure 54: Population trajectories across the stated sensitivity models.

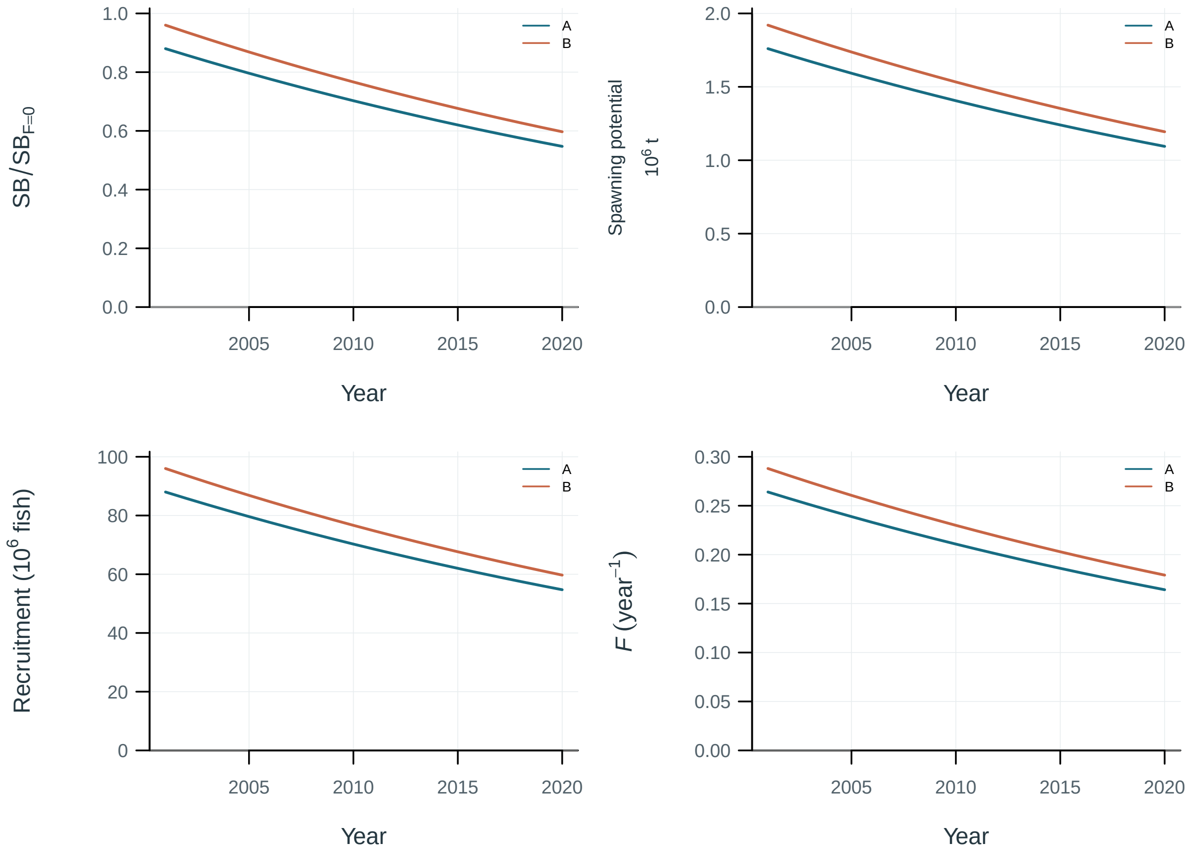

Figure 55: Population trajectories across the stated sensitivity models.

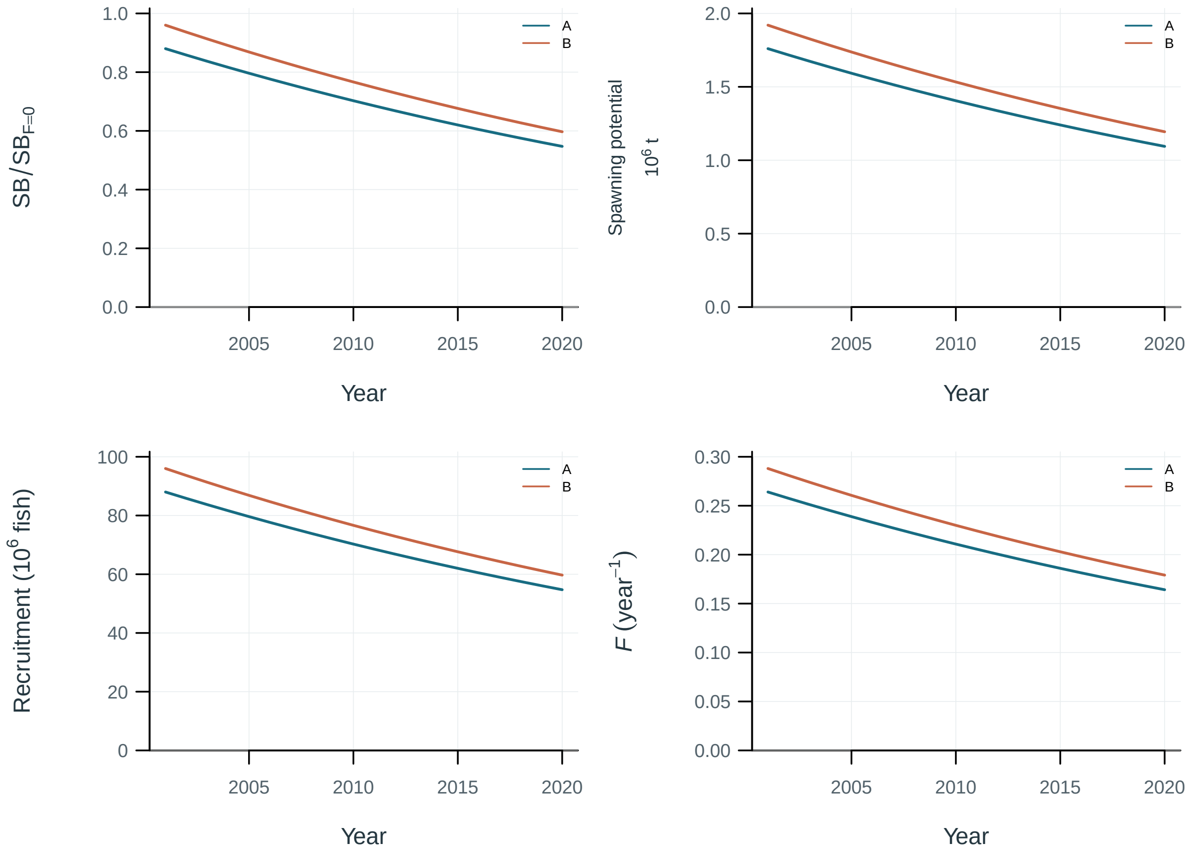

Figure 56: Population trajectories across the stated sensitivity models.

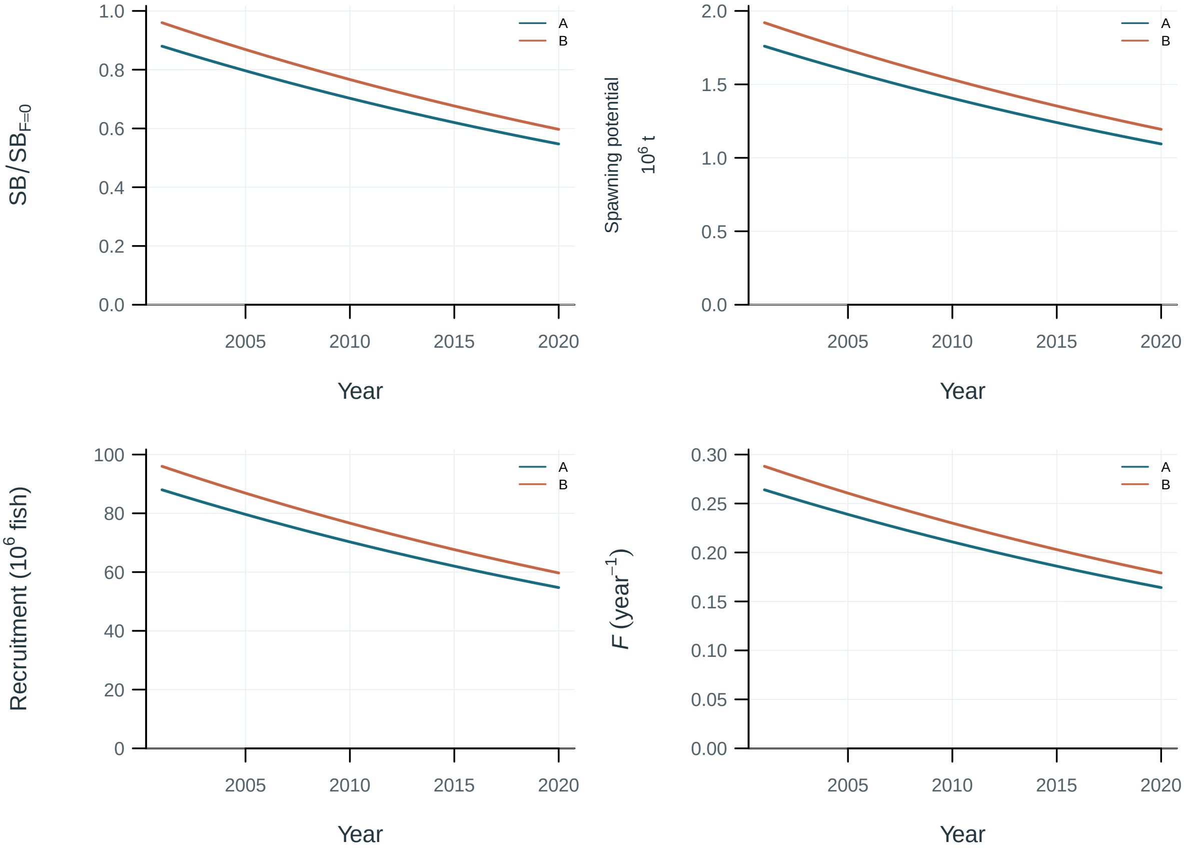

Figure 57: Population trajectories across the stated sensitivity models.

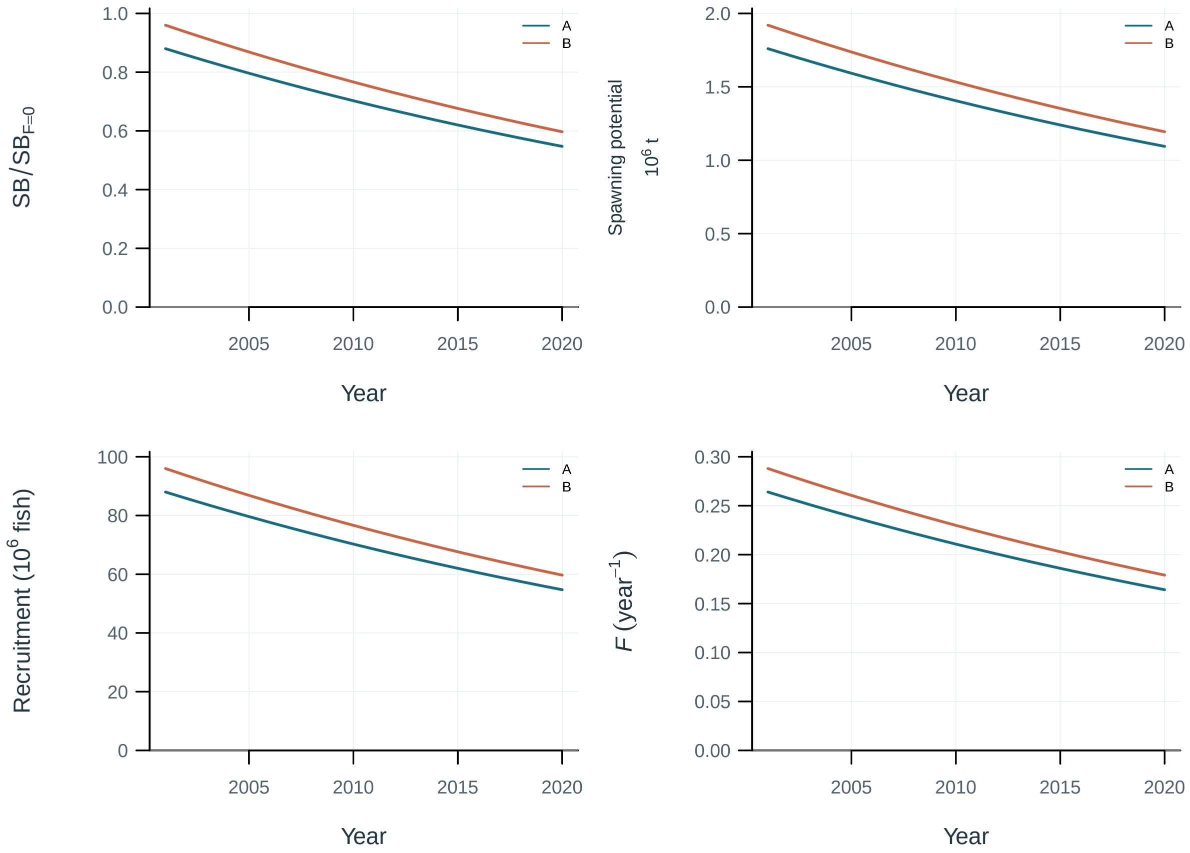

Figure 58: Population trajectories across the stated sensitivity models.

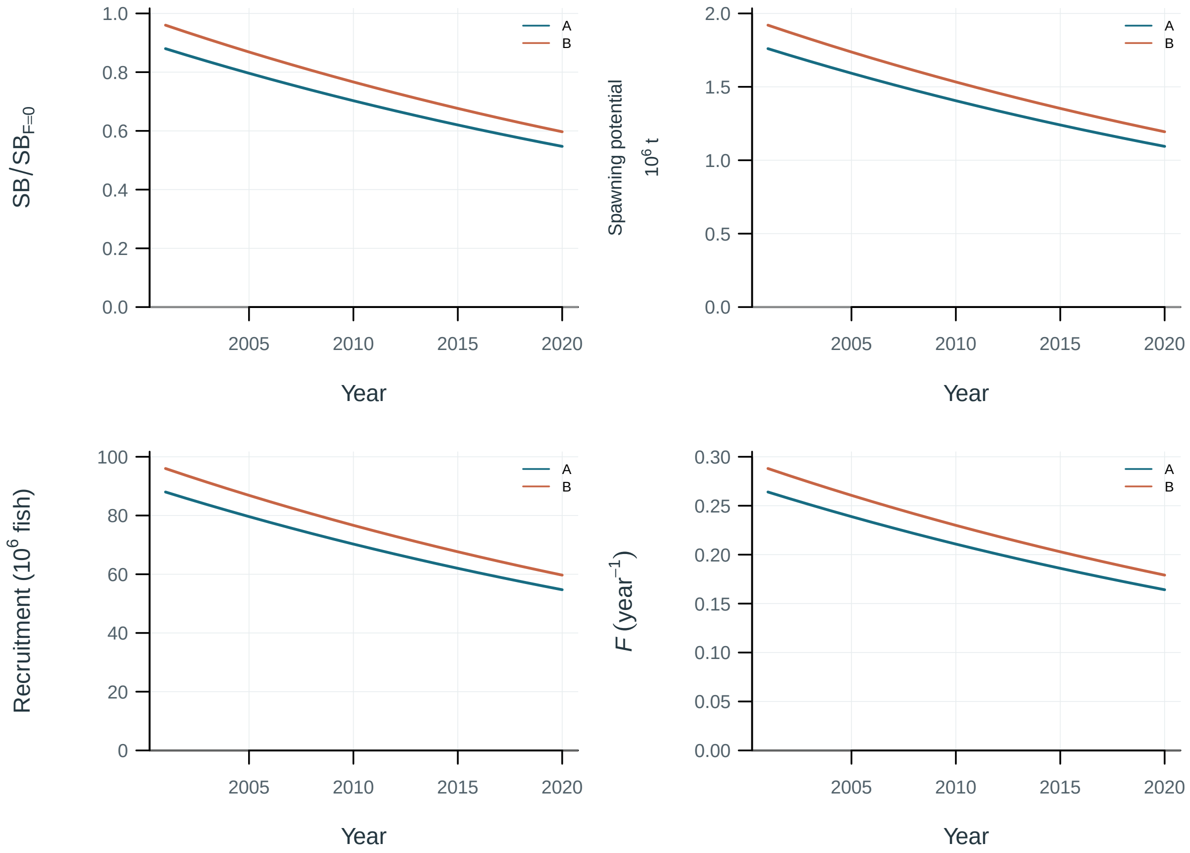

Figure 59: Population trajectories across the stated sensitivity models.

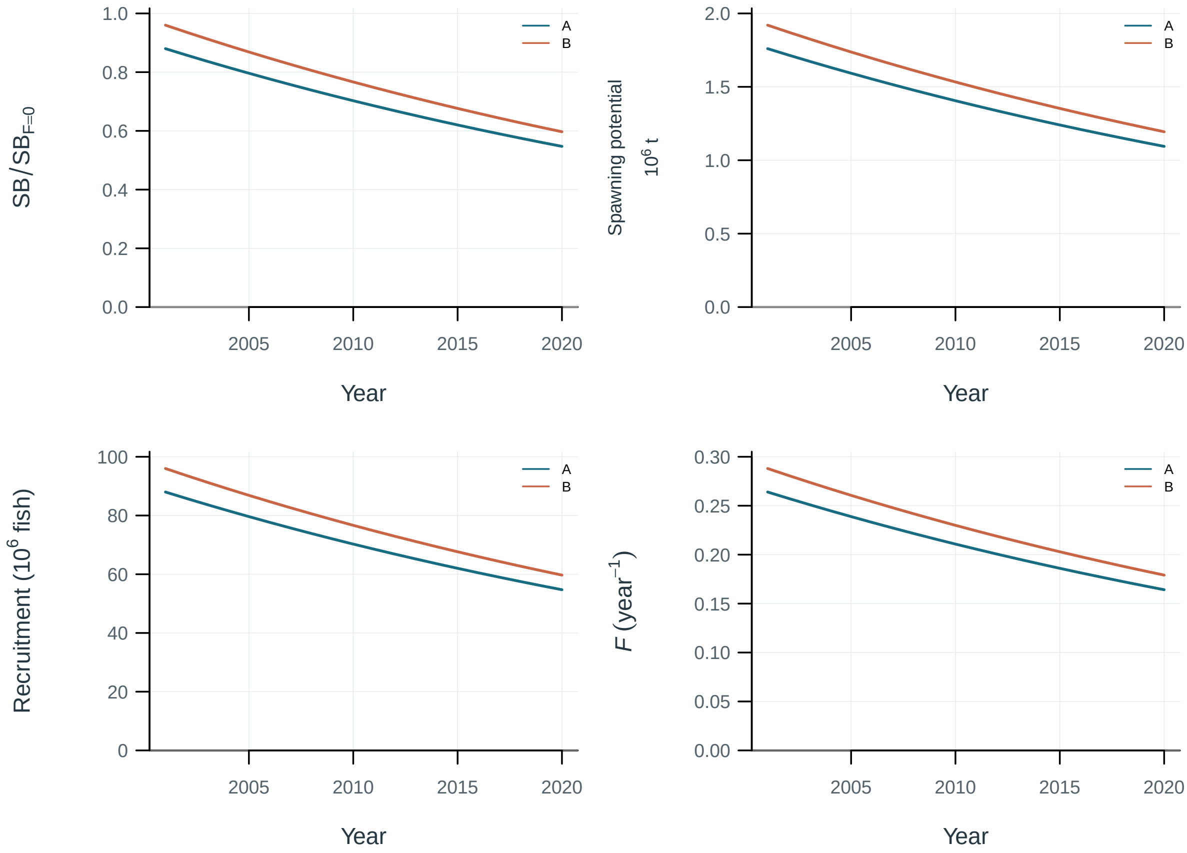

Figure 60: Population trajectories across the stated sensitivity models.

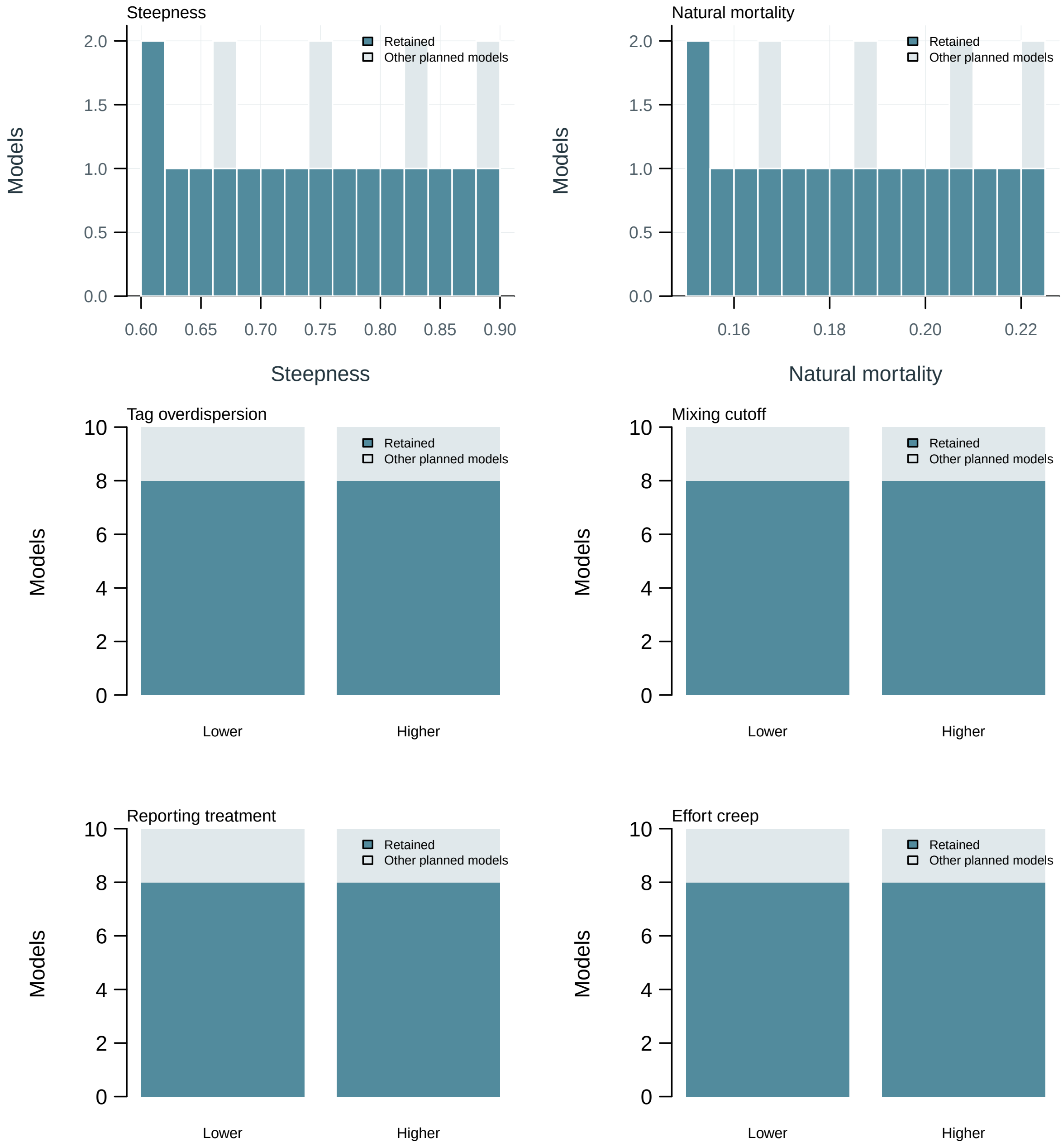

Figure 61: Planned and retained model inputs.

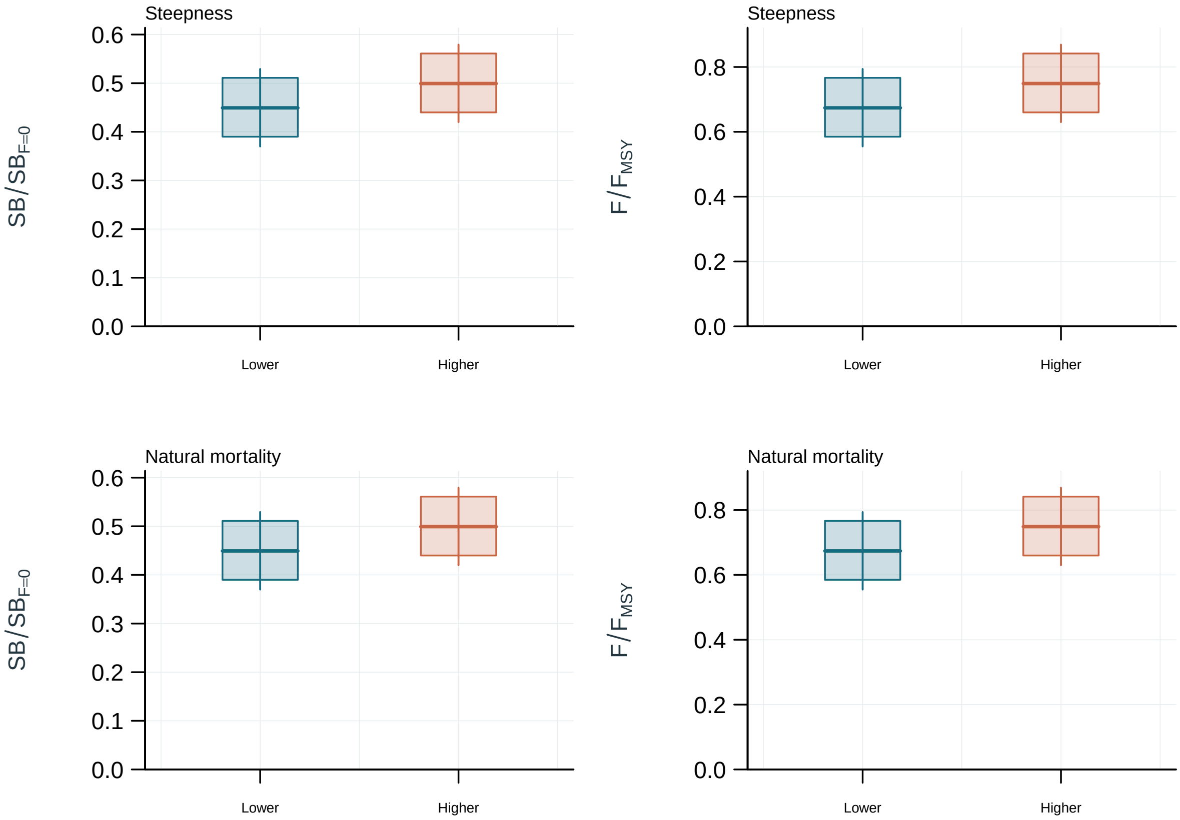

Figure 62: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

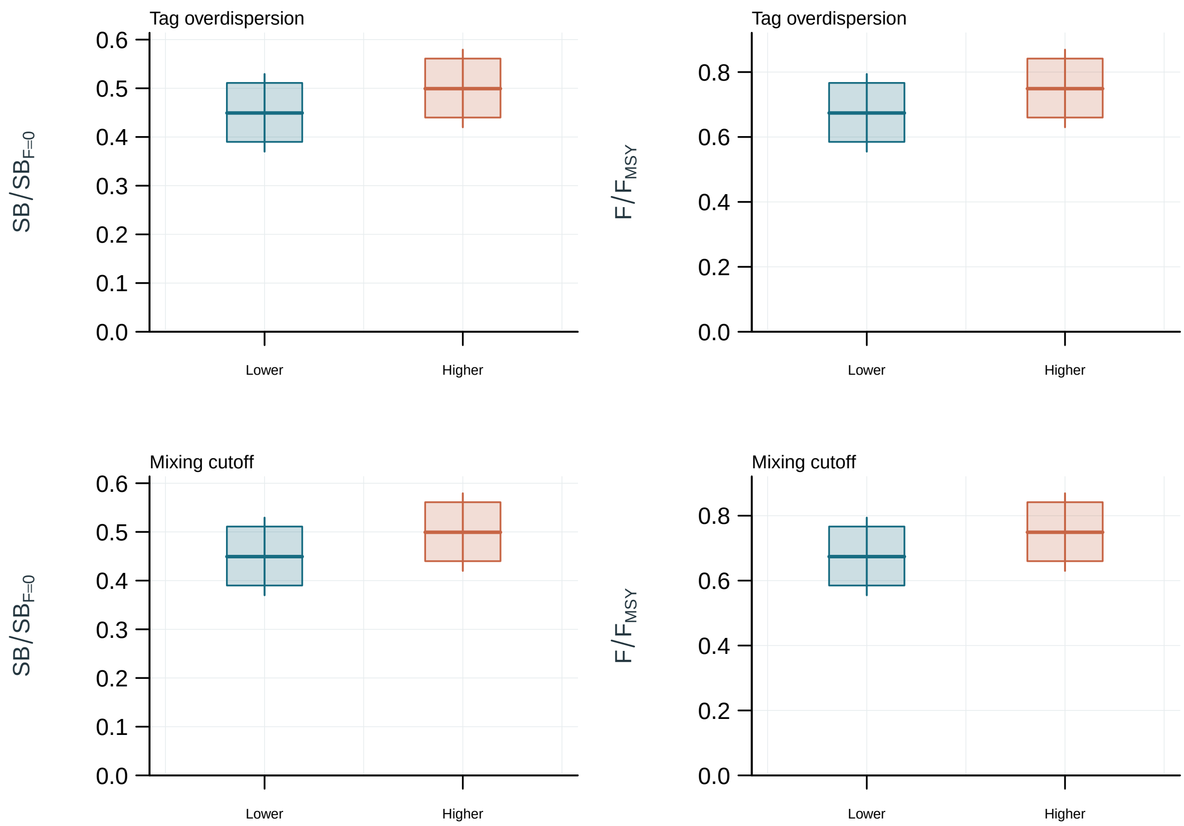

Figure 63: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

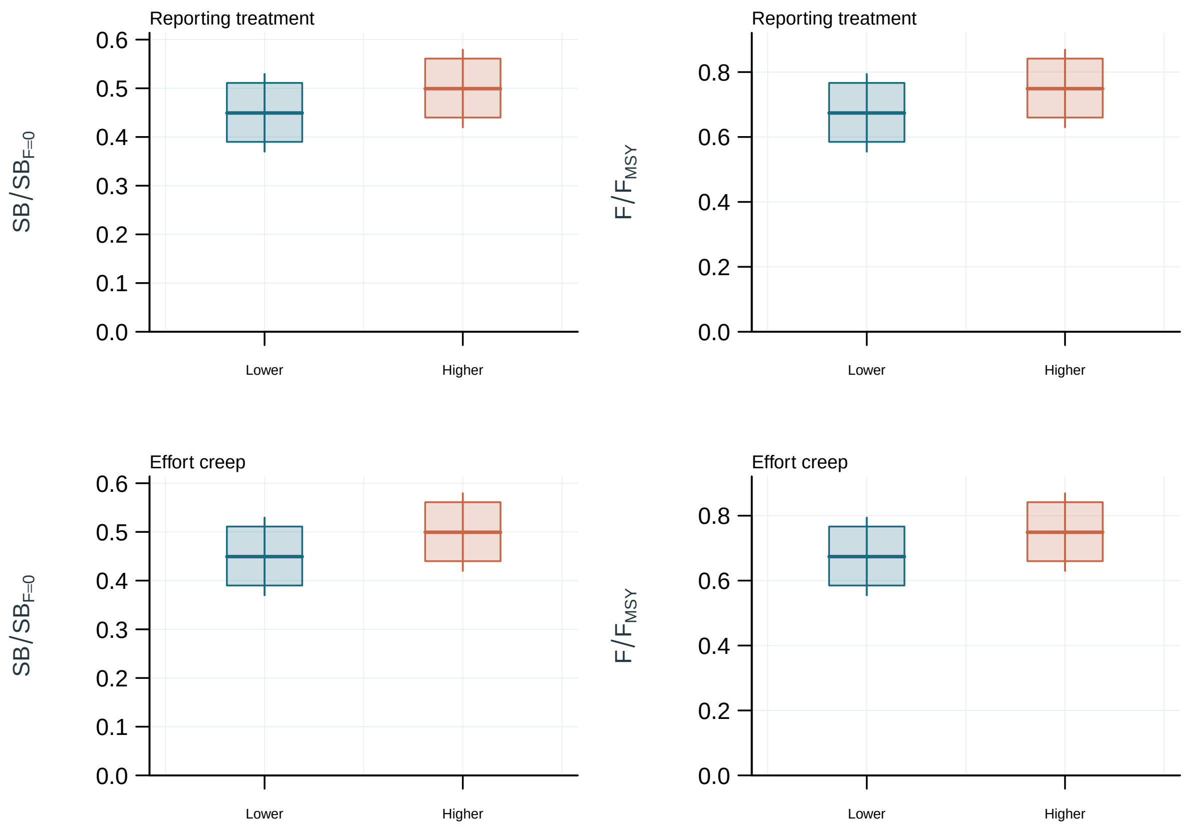

Figure 64: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

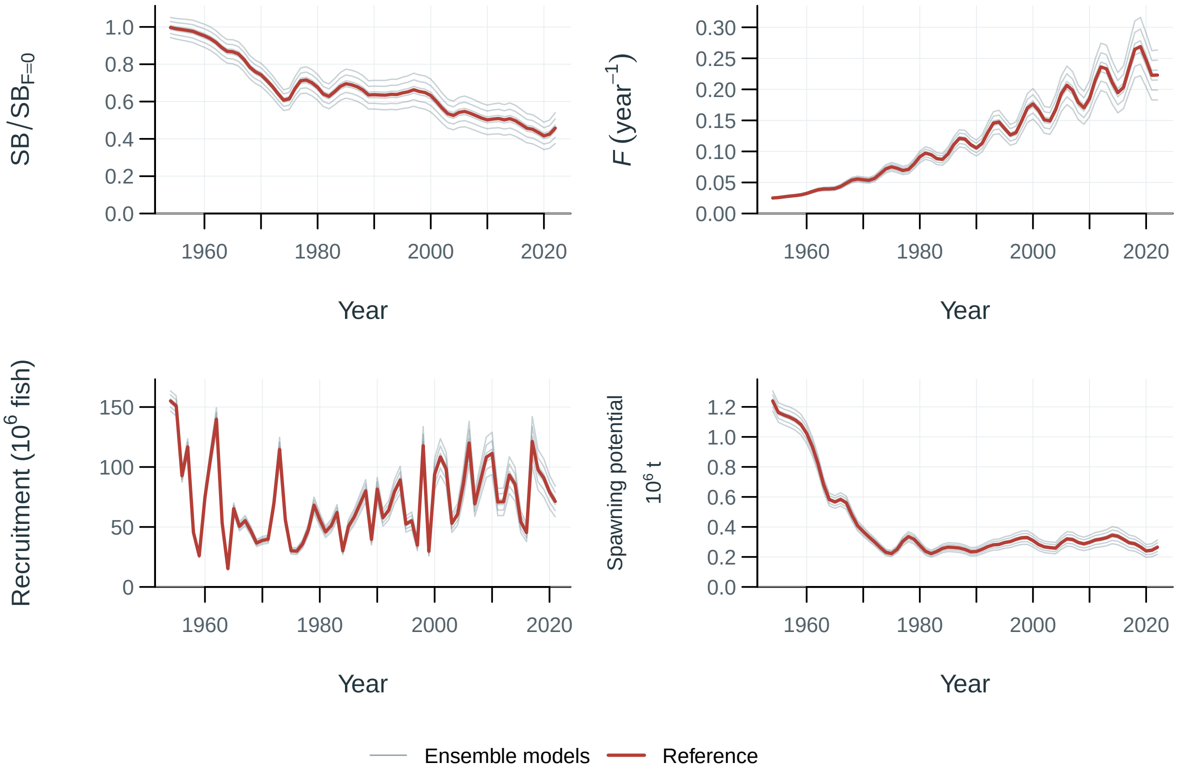

Figure 65: Population trajectories across ensemble models (thin lines) and the reference model (red).

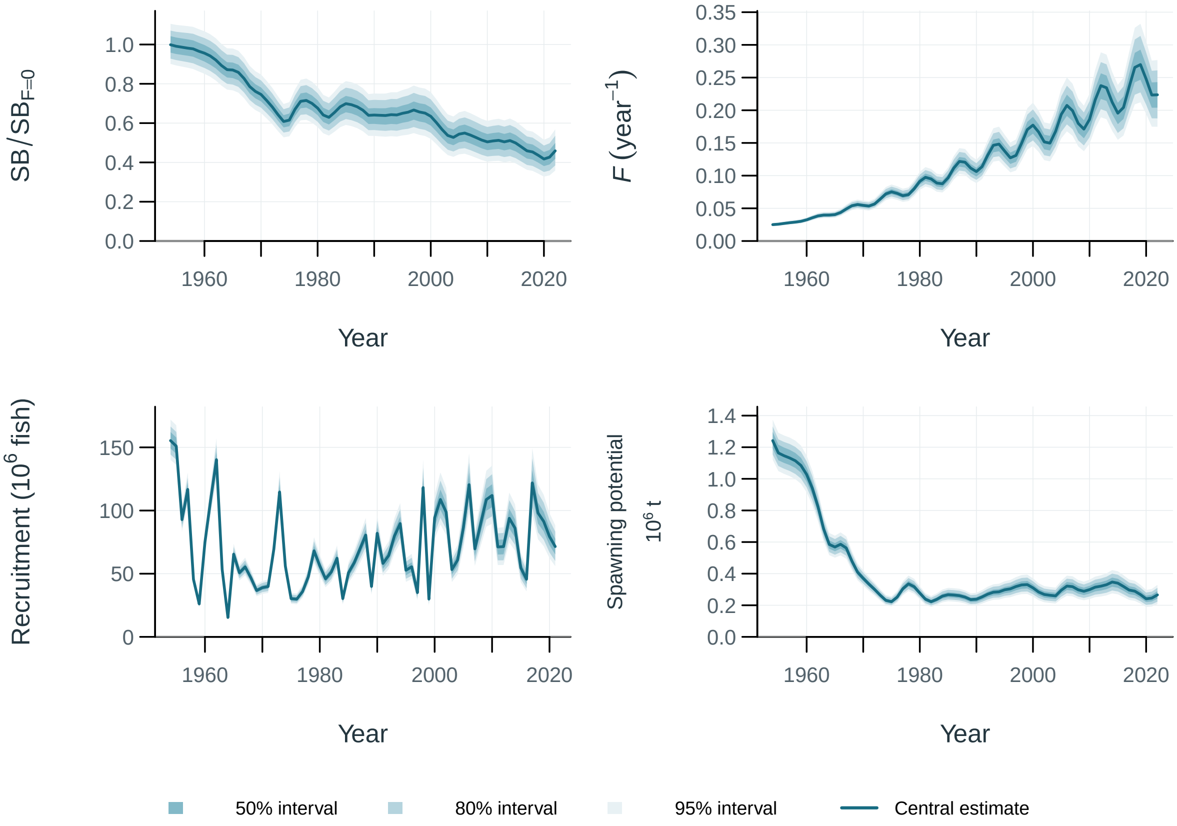

Figure 66: Population trajectories with supplied uncertainty intervals. Central 50, 80, 95% intervals (illustrative model spread and parameter perturbations).

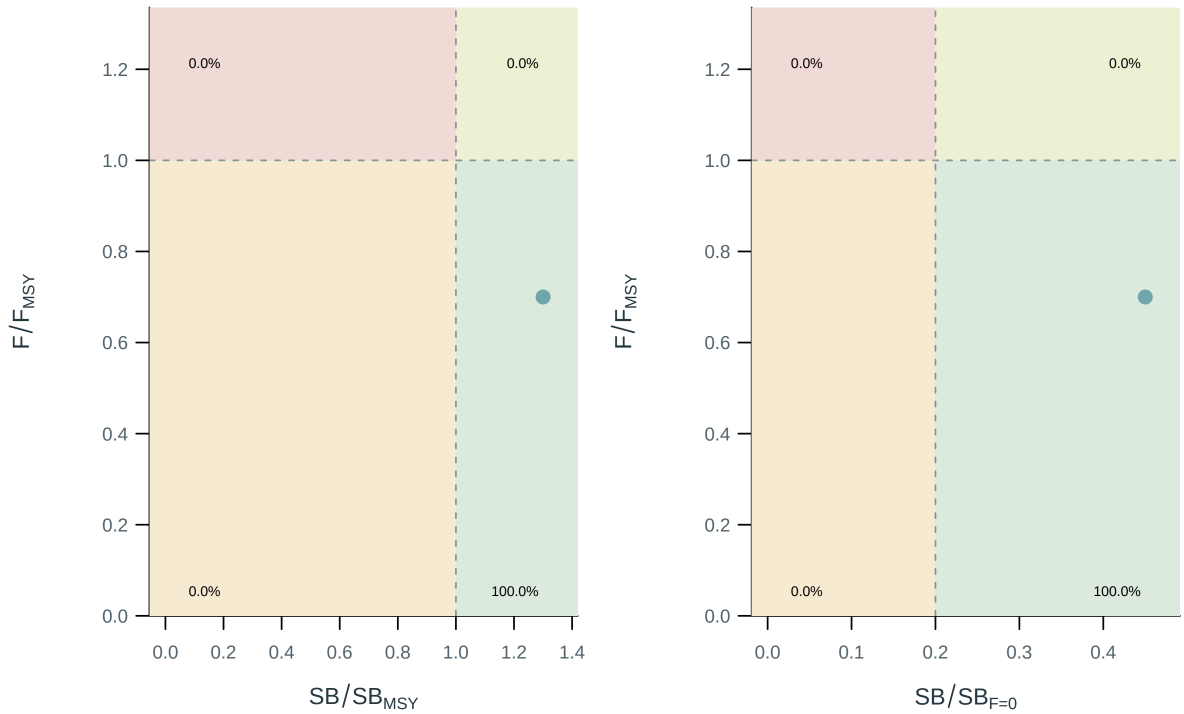

Figure 67: Stock status relative to the supplied spawning and fishing reference points.

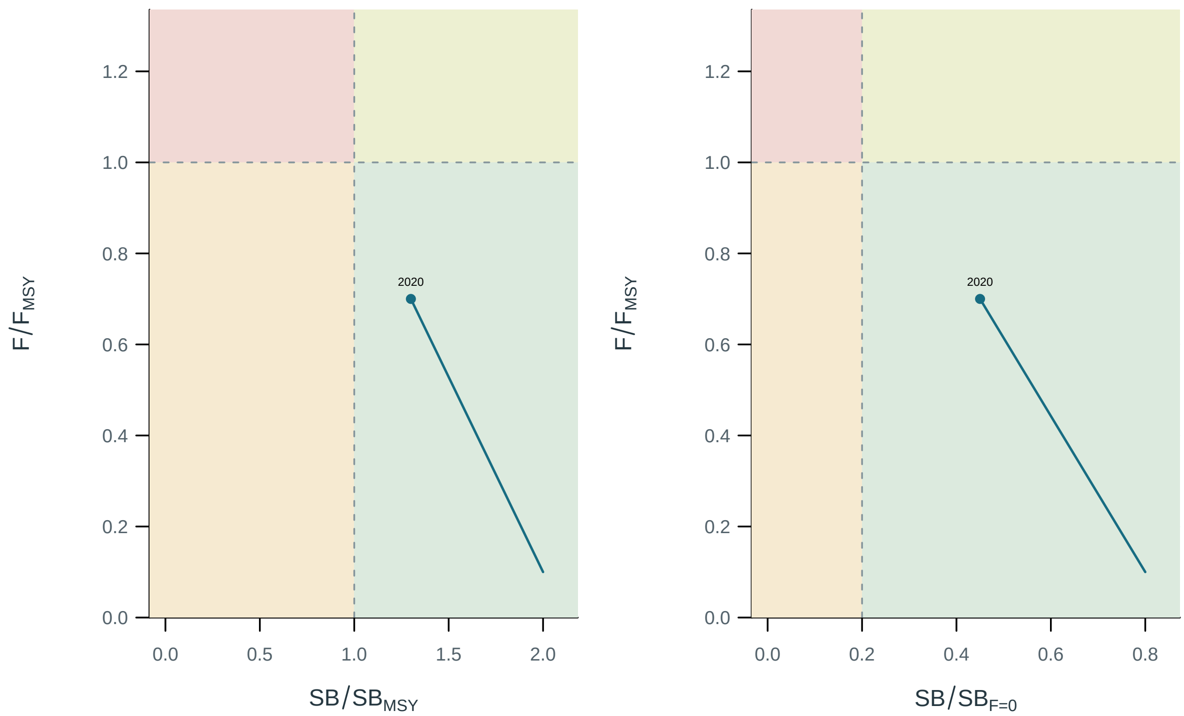

Figure 68: Stock status relative to the supplied spawning and fishing reference points.

Figure 69: Dynamic maximum sustainable yield and catch.

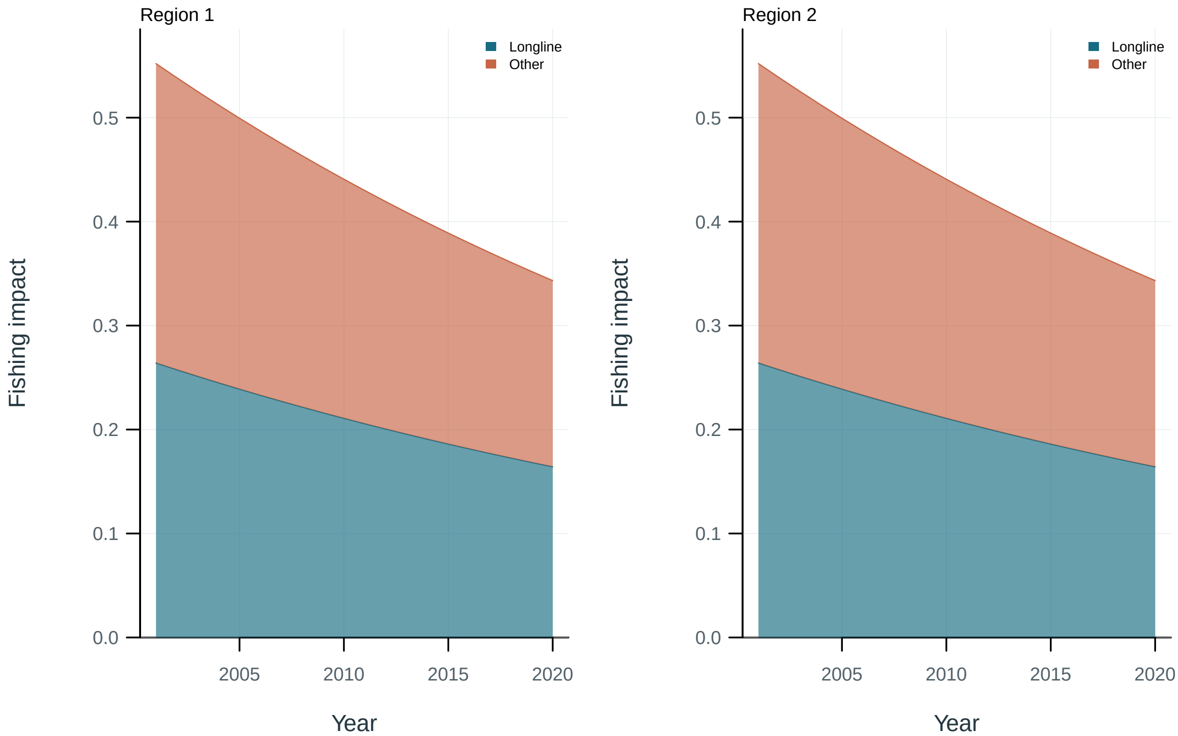

Figure 70: Illustrative output. Contributions of fishing groups to reduced spawning potential.

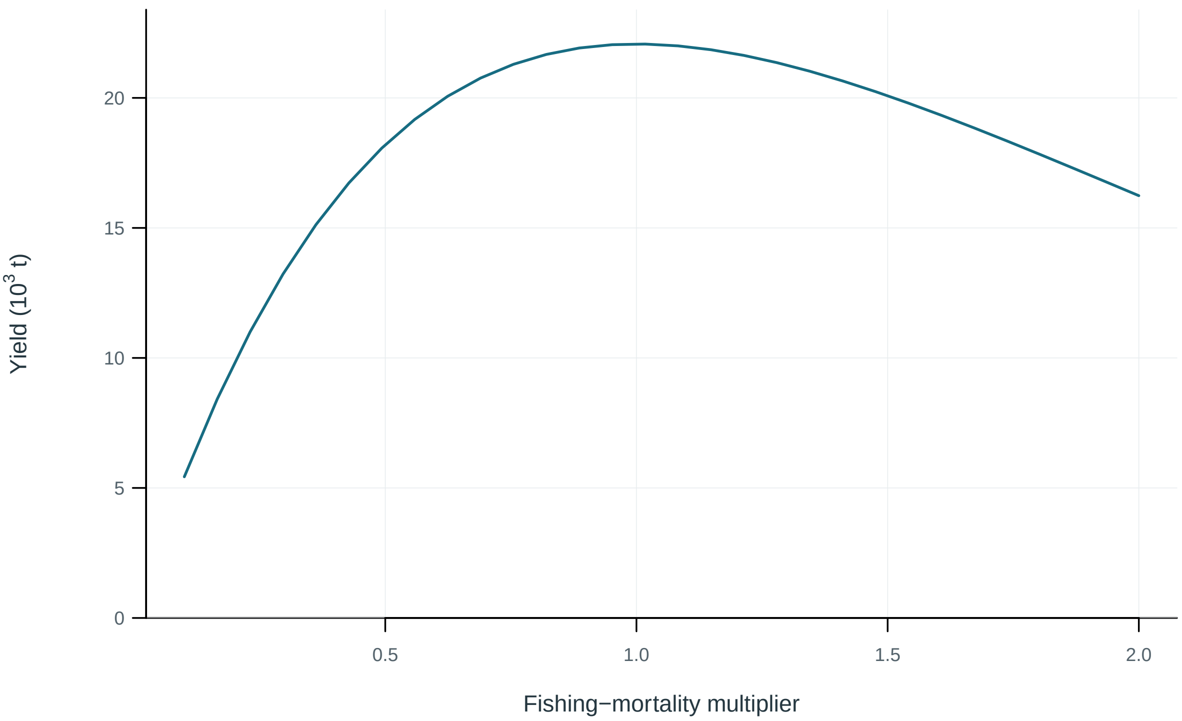

Figure 71: Equilibrium yield across fishing-mortality multipliers.

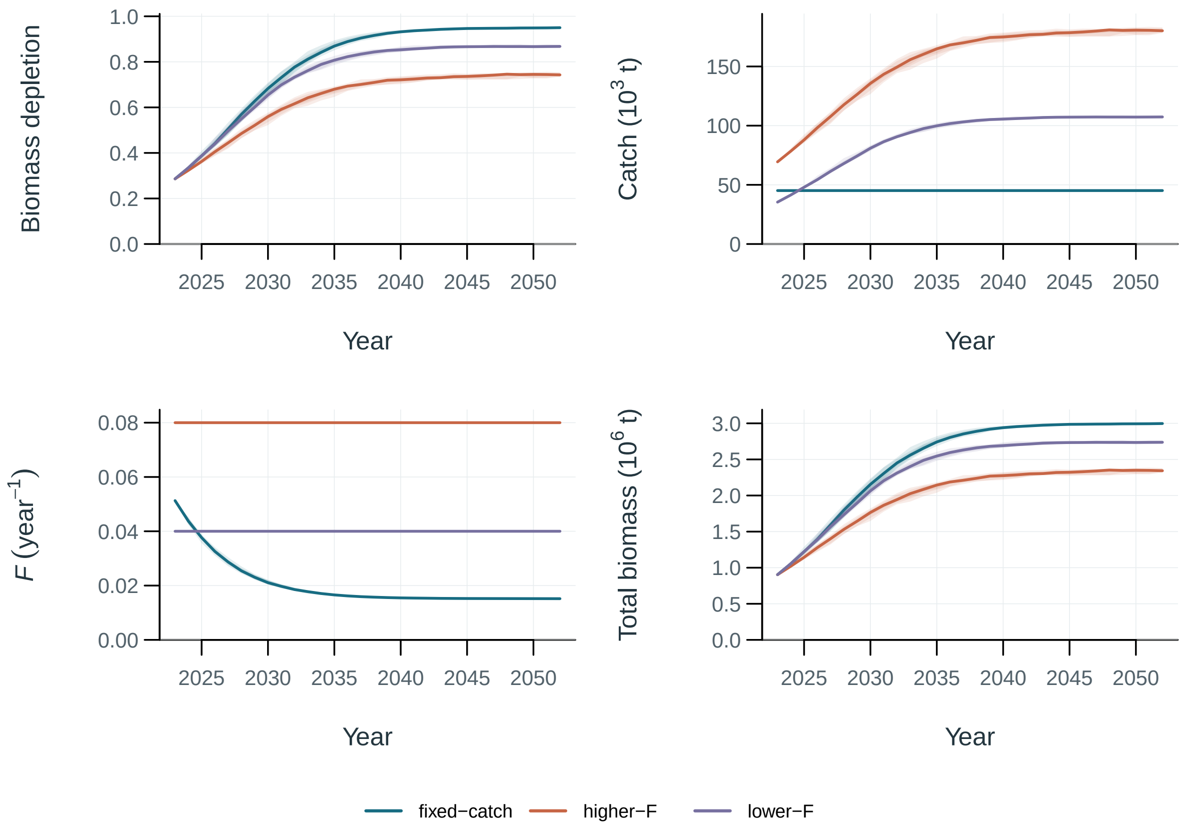

Figure 72: Illustrative surplus-production projections under fixed catch and two F scenarios. Lines are medians; bands span the central 50%, 80% and 95% of production-variability replicates.

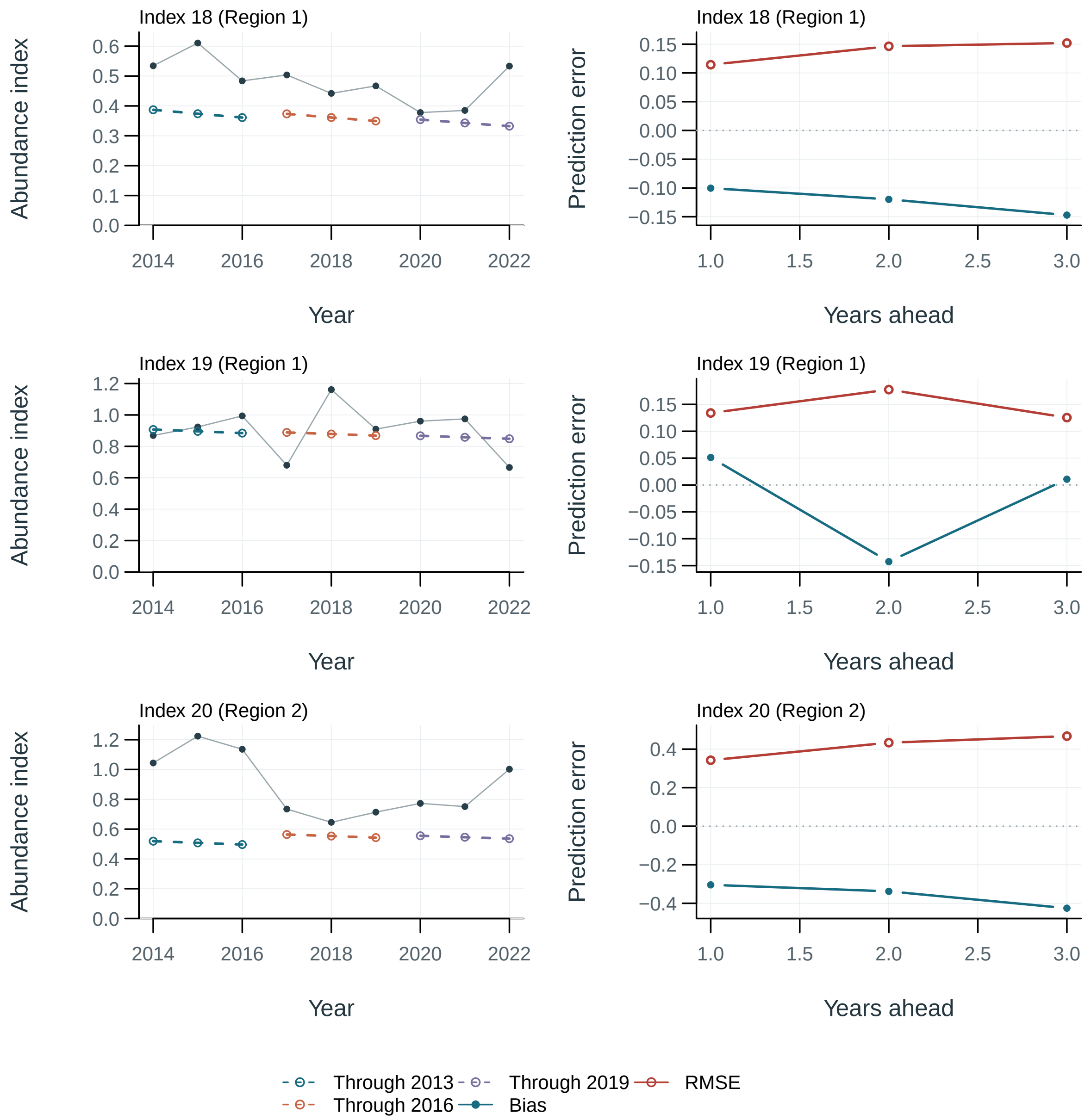

Figure 73: Withheld observations (filled points) and predictions from successive training cutoffs (dashed lines). Right panels show prediction bias and RMSE by forecast lead, in observation units.

11 Supplementary outputs

ALB · Diagnostic model interactive comparison

ALB · Diagnostic model figures and tables

ALB · Native MFCL and KIMSA interactive comparison

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ALB · Starting-value trials interactive comparison

ALB · Starting-value trials figures and tables

ALB · Self-test recovery interactive comparison

ALB · Self-test recovery figures and tables

ALB · Retrospective analysis interactive comparison

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ALB · Sensitivities interactive comparison

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ALB · Uncertainty ensemble interactive comparison

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Hindcasting interactive comparison

Hindcasting figures and tables

Projection scenarios interactive comparison

Projection scenarios figures and tables

12 Appendix: supplementary methods

13 References